Friction :-

* friction is due to interlocking of irregularities of surfaces by molecular bondings.
* Friction always acts "tangential" to the surfaces in contact.
* It always "opposes" the "relative motion" b/w the surfaces in contact.
* while walking, friction on man acts in the "direction of walking".
* Friction always forms "action-reaction" pair

Types of friction:

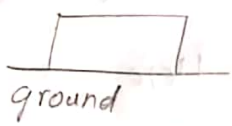
1. static friction
2. Kinetic / Dynamic friction
3. Rolling friction

Static friction:

* It acts b/w the surfaces in contact only when there is no "relative motion" b/w them.
* Static friction is a self adjusting force. $f_s = F(\text{applied})$
* Maximum value of static friction is called "limiting friction".

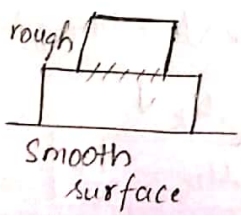
$$(f_s)_{max} = f_l$$

* $(f_s)_{max} = F$.



* Limiting friction is always directly proportional to Normal reaction. i.e, $(f_s)_{max} \propto N$ ⟶ laws of motion.

$$(f_s)_{max} = \mu_s N$$



$$\frac{f_l}{N} = \mu_s = \text{coefficient of static friction.}$$

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$0 < \mu_s < 1$

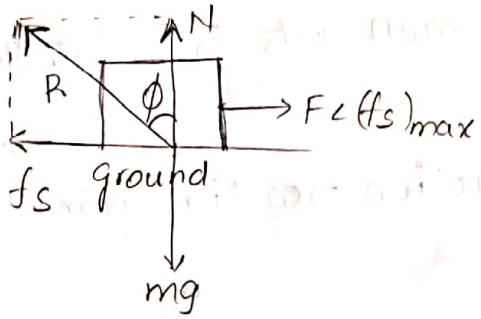
* when surface is tomuch polished then $\mu_s > 0$.

So $\mu > 1$ for highly polished surfaces.

$\mu = 0$ for smooth surfaces.

Contact force

*



Here body is at rest

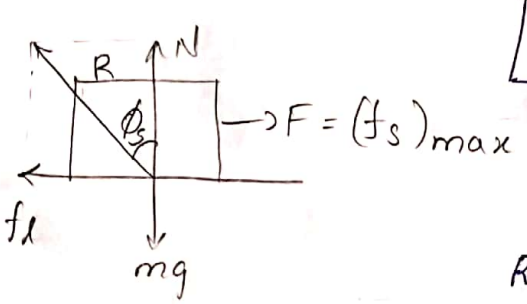
$N = mg$

$F = f_s$

$R = \sqrt{N^2 + f_s^2}$

$\boxed{R = \sqrt{N^2 + F^2}}$

*



$f_l = \mu_s N$ [when $F = f_l$ then the body just begins to slide]

$R = \sqrt{N^2 + f_l^2}$

$= \sqrt{N^2 + \mu_s^2 N^2}$

$R = N\sqrt{1 + \mu_s^2}$

$\boxed{R = mg\sqrt{1 + \mu_s^2}}$

max angle $= \phi_s$

= angle of friction.

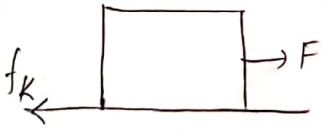
$\tan \phi_s = \frac{f_l}{N} = \frac{\mu_s N}{N}$

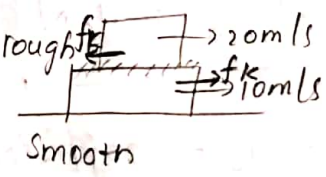
$\boxed{\tan \phi_s = \mu_s} \Rightarrow \boxed{\phi_s = \tan^{-1}(\mu_s)}$

Kinetic friction:

* Here $F > f_l$, then body slides over another body.

* Here kinetic friction decreases and opposes the "relative velocity" b/w the surfaces in contact.





$f_k \propto N \rightarrow$ law of

$f_k = \mu_k N$

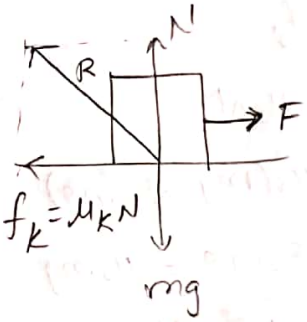
$\boxed{\mu_k = \frac{f_k}{N}}$

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* usually, $f_k < f_s \Rightarrow \boxed{\mu_k < \mu_s}$

* f_k is independent of relative velocity.

Contact force

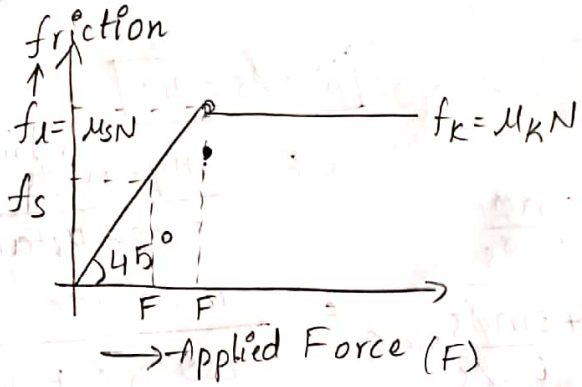


$$R = \sqrt{N^2 + f_k^2}$$
$$= \sqrt{N^2 + \mu_k^2 N^2}$$
$$\boxed{R = N\sqrt{1+\mu_k^2}}$$

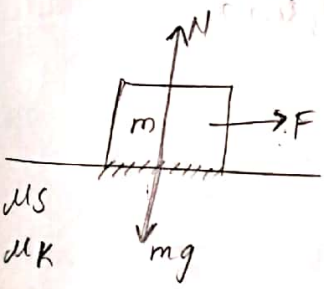
$$\tan\phi = \frac{f_k}{N} = \frac{\mu_k N}{N} \Rightarrow \boxed{\tan\phi = \mu_k}$$

Graph :-





*



① $N = mg$

② $f_s = \mu_s N = \mu_s mg$

③ $f_k = \mu_k N = \mu_k mg$

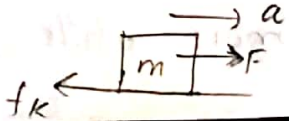
*If $F < f_s$; block is at rest.

friction → static (f_s) Here $\boxed{f_s = F}$

Eg If $\boxed{F = f_s}$; Body ready to slide.

↳ min force required to move the body.

* Even after the motion has started, if same force is continued to act on the body,



$$a = \frac{f_s - f_k}{m} = \frac{\mu_s N - \mu_k N}{m}$$

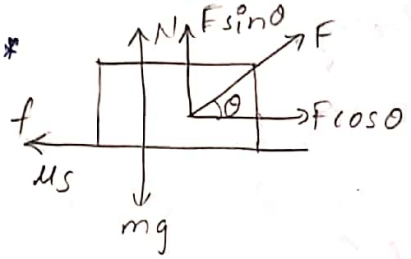
$$\Rightarrow \boxed{a = (\mu_s - \mu_k)g}$$

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Inclined pulling force :-

* min force required to pull the block :-

*



$N = mg - F\sin\theta$ — (1)

* just to pull the block,

$F\cos\theta = f_l$

$F\cos\theta = \mu_s N$

$F\cos\theta = \mu_s(mg - F\sin\theta)$

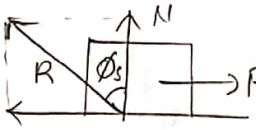
$F\cos\theta + \mu_s F\sin\theta = \mu_s mg$

$F(\cos\theta + \mu_s \sin\theta) = mg\mu_s$

$$F = \frac{\mu_s mg}{\cos\theta + \mu_s \sin\theta}$$

↳ min force required

*



$\tan\phi_s = \mu_s$

$$F = \frac{mg\tan\phi_s}{\cos\theta + \tan\phi_s \sin\theta}$$

$$F = \frac{mg\frac{\sin\phi_s}{\cos\phi_s}}{\cos\theta + \frac{\sin\phi_s}{\cos\phi_s}\sin\theta} = \frac{mg\sin\phi_s}{\cos\theta\cos\phi_s + \sin\phi_s\sin\theta}$$

$$F = \frac{mg\sin\phi_s}{\cos(\theta - \phi_s)}$$ → also min force.

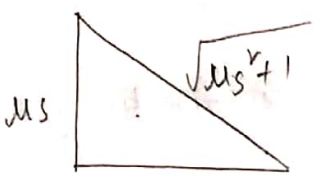
* for least value of min force.

$\cos(\theta - \phi_s) = 1$

$\theta - \phi_s = 0 \Rightarrow \theta = \phi_s$

$F_{least} = mg\sin\phi_s$

*

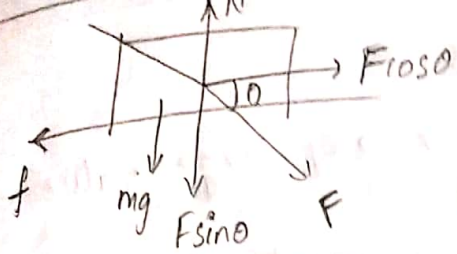


$$F_{least} = \frac{\mu_s mg}{\sqrt{1+\mu_s^2}}$$

Inclined pushing force :-

* friction developed is more while pushing.

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$$N = mg + F\sin\theta \quad -(1)$$

* just to move the block,

$$F\cos\theta = f_s$$

$$F\cos\theta = \mu_s N$$

$$F\cos\theta = \mu_s (mg + F\sin\theta)$$

$$F(\cos\theta - \mu_s \sin\theta) = \mu_s mg$$

$$\boxed{F = \frac{\mu_s mg}{\cos\theta - \mu_s \sin\theta}}$$

min pushing force,

$$\boxed{F = \frac{\mu_s mg}{\cos\theta - \mu_s \sin\theta}}$$

*we have, $\tan\phi_s = \mu_s$

$$F = \frac{\tan\phi_s\, mg}{\cos\theta - \tan\phi_s \sin\theta} = \frac{\frac{\sin\phi_s}{\cos\phi_s} mg}{\cos\theta - \frac{\sin\phi_s}{\cos\phi_s}\sin\theta}$$

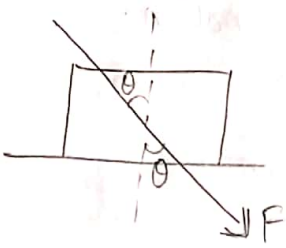
$$\boxed{F = \frac{\sin\phi_s\, mg}{\cos(\theta + \phi_s)}} \rightarrow \text{min force.}$$

* for least value of min force,

$\theta = 0°$, $\phi_s \rightarrow$ const.

$$\boxed{F_{least} = mg\tan\phi_s = mg\mu_s}$$

Very Imp point:



If F is making an angle 'θ' with vertical, $\theta \rightarrow 90-\theta$

min force $F = \dfrac{\mu_s mg}{\sin\theta - \mu_s\cos\theta}$

For the body to slide, $F = +ve$

$$\sin\theta - \mu_s\cos\theta > 0$$

$$\tan\theta > \mu_s$$

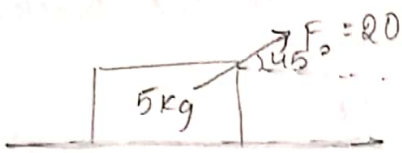
$$\tan\theta > \tan\phi_s$$

$$\boxed{\theta > \phi_s} \rightarrow F = +ve.$$

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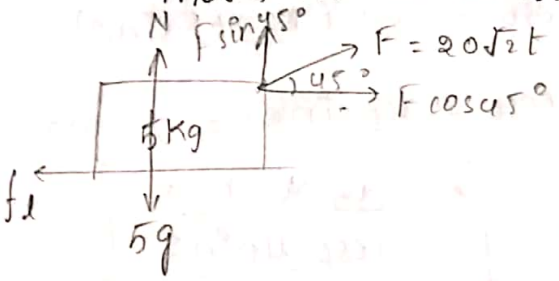
* If $\theta < \phi_s$ (or) $\tan\theta < \mu_s$

=) block will not slide for any large value of 'F'



$\mu_s = \mu_k = 0.2$

Find
① the moment when block leaves the contact.
② the moment when block just moves in horizontal direction.



$N - 5g = F\sin 45^\circ$

$N - 5g = 20\sqrt{2}t \cdot \frac{1}{\sqrt{2}}$

$N - 5g = 20t$

① $N = 0$

$5g = 20t \Rightarrow \boxed{t = 2.5 \text{ sec}}$

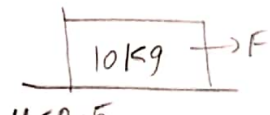
② $F\cos 45^\circ = f_l$

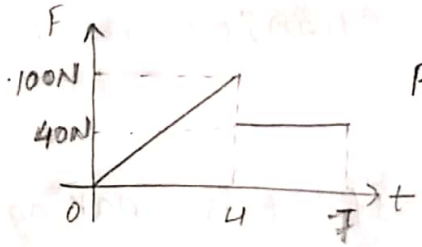
$20\sqrt{2}t \cdot \frac{1}{\sqrt{2}} = \mu_s (5g - 20t)$

$20t = 0.2 (50 - 20t)$

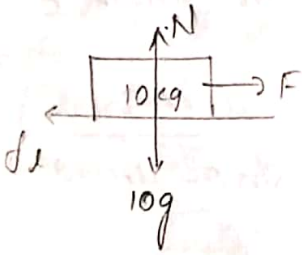
$10t = 5 - 2t$

$12t = 5 \Rightarrow \boxed{t = 5/12 \text{ sec}}$





Find total time during which block is in motion.



$F = f_l = \mu_s N = \mu_s mg$

$F = 0.5 \times 10 \times 10$

$\boxed{F = 50N}$

* from graph,

$F = \left(\frac{100}{4}\right) t = 25t$

Block just begines to slide, $50 = \left(\frac{50}{2}\right) t \Rightarrow \boxed{t = 2 \text{sec}}$

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* upto 2 sec, block is at rest.

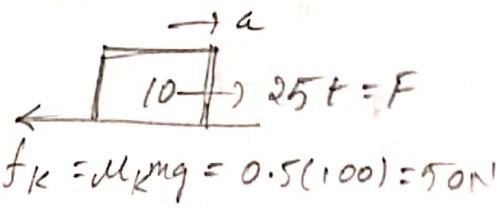
$\therefore$ friction is static.

$f_s = F = 25t \quad (0 \le t \le 2)$

i.e, At $t = 1$ sec, $f_s = 25(1) = 25N$.

$t = 1.5$, $f_s = 37.5N$.

* between $t = 2$ sec & 4 sec.



$$a = \frac{25t - 50}{10}$$

$$a = 2.5t - 5$$

$$\frac{dv}{dt} = 2.5t - 5$$

$$\int_0^v dv = \int_2^t (2.5t - 5)\,dt$$

$$v = \left[\frac{2.5t^2}{2} - 5t\right]_2^t$$

$$v = \frac{2.5(t^2 - 4)}{2} - 5(t - 2)$$

$$v = \frac{2.5t^2}{2} - 5 - 5t + 10$$

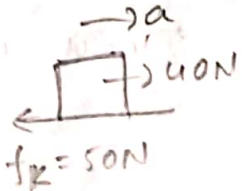
$$v = \frac{2.5t^2}{2} - 5t + 5$$

$$\boxed{v = 1.25t^2 - 5t + 5} \quad (\text{for } 2 \le t \le 4)$$

$v_{max} \Rightarrow v = 1.25(4)^2 - 5(4) + 5$

at $t = 4$ sec $= 20 - 20 + 5 = 5$ m/s

now for $4 \le t \le 7 \Rightarrow F = 40N$



$$a = \frac{40 - 50}{10} = -1\ m/s^2$$

a → backward $\Rightarrow$ v is decreasing.

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$$\frac{dv}{dt} = -1$$

$$\int_5^v dv = -\int_4^t 1\,dt$$

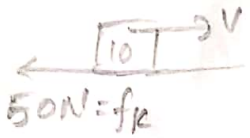
$$v - 5 = 4 - t$$

$$\boxed{v = 9 - t} \quad (\text{for } 4 \le t \le 7)$$

$\therefore$ velocity at $t = 7\,\text{sec}$, $v = 2\,\text{m/s}$

---

after $t = 7$ s: $a = \frac{-50}{10}$



$$a = -5\,\text{m/s}^2$$

$$\frac{dv}{dt} = -5$$

$$\int_2^v dv = -\int_7^t 5\,dt$$

$$v - 2 = 35 - 5t$$

$$\boxed{v = 37 - 5t} \quad (\text{for } t \ge 7)$$

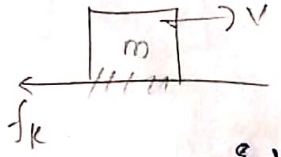
$v = 0$ when $5t = 37$

$$\boxed{t = 37/5}$$

$$\boxed{t = 7.4\,\text{sec}}$$

This is the total time from $t = 0$.

$\therefore$ Total time during motion $= 7.4 - 2 = 5.4\,\text{sec}$.



$$a_m = \frac{-f_k}{m} = \frac{-\mu_k mg}{m} = -\mu_k g$$

& $v = u + at \Rightarrow v = v_i + at$

$v = 0 \Rightarrow v_i = -at$

$v_i = \mu_k g t$

$$\boxed{t = \frac{v_i}{\mu_k g}}$$

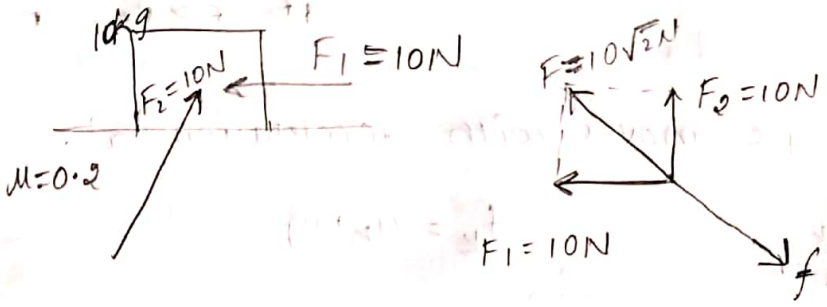
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$v^2 - u^2 = 2as$

$0 - v_1^2 = 2(-\mu_k g)(s)$

$$\boxed{\frac{v_1^2}{2\mu_k g} = s} \rightarrow \text{stopping distance.}$$

→



$f_d = \mu_s N$

$= 0.2 \times 100$

$f_d = 20N$

Here $F < f_d$ → block is at rest

⇒ static friction. $f_s = F = 10\sqrt{2}\,N$.

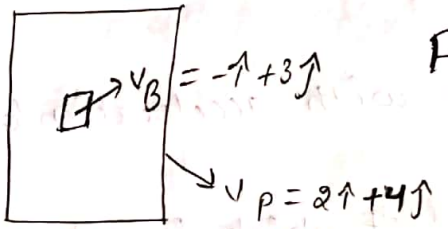
* If $\mu = 0.1$, $f_d = 10N$

$F > f_d$ → block slides.

⇒ friction is kinetic, $f_k = 10\,N$.

$$a = \frac{10\sqrt{2} - 10}{10} = \sqrt{2} - 1 = 0.414\,m/s^2.$$

→



Find unit vector in the direction of friction.

$\vec{v}_{b/p} = \vec{v}_b - \vec{v}_p$

$= -\hat{i} + 3\hat{j} - 2\hat{i} - 4\hat{j}$

$$\boxed{\vec{v}_{b/p} = -3\hat{i} - \hat{j}}$$

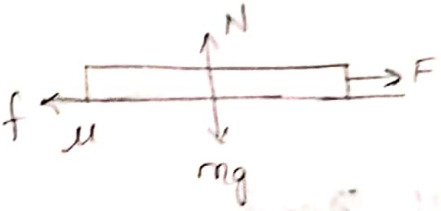
Vector along friction,

$$\hat{n} = \frac{-(\vec{v}_{b/p})}{|\vec{v}_{b/p}|}$$

$$= \frac{-(-3\hat{i} - \hat{j})}{\sqrt{10}} = \frac{3\hat{i} + \hat{j}}{\sqrt{10}}$$

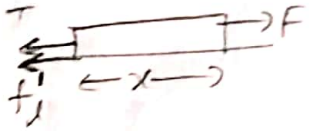
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Tension at a point in the rope:-



$f_l = \mu_s N = \mu mg$

* If $F < f_l$, rope is at rest.



$f_l' = \mu \frac{m}{l} g$

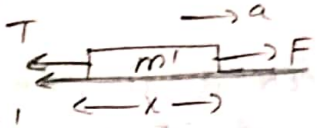
if $F < F_l'$ (Max) then $F_l' = F$; $T = 0$

if $F > F_l'$, $T = F - F_l'$

* If $F > f_l$, rope moves with acceleration 'a'.

$a = \frac{F - f_k}{m}$

$f_k = \mu_k mg$



$f_k' = \mu_k \times \frac{m}{l} \times g$

$F - T - f_k' = m'a$

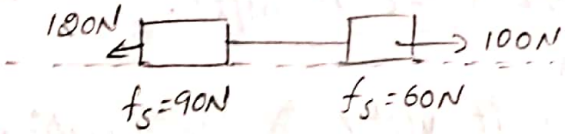
$T = F - f_k' - m'a$

$\boxed{T = F - f_k' - \frac{m}{l} g}$

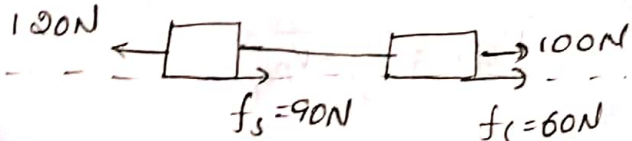


Connecting bodies:-

(1) Find tension in string.

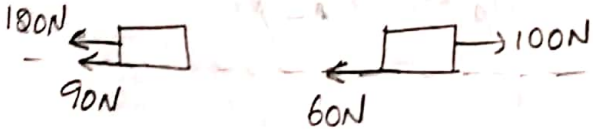


(i) The system moves to left with acceleration 'a'.



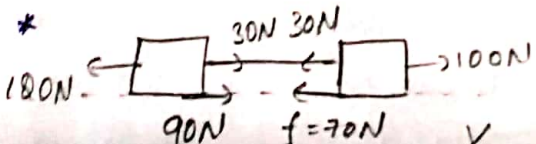
which is not possible

(ii) The system moves to right with acceleration 'a'.

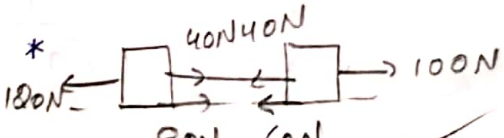


which is not possible.

∴ System is at rest.

* 

✗

* 

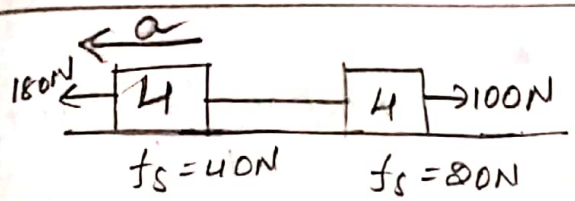
✓

∴ $\boxed{T = 40N}$

$f = 60N (\leftarrow)$
$= 80N (\rightarrow)$

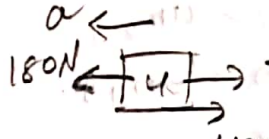
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→



$$a = \frac{180-100-60}{8} = \frac{20}{8} = \frac{5}{2}$$

$$\boxed{a = 2.5 m/s^2}$$

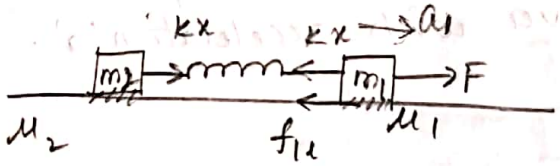


$$180 - T - 40 = 4a$$

$$180 - 40 - 10 = T$$

$$\boxed{T = 130N}$$

* Connected with spring:

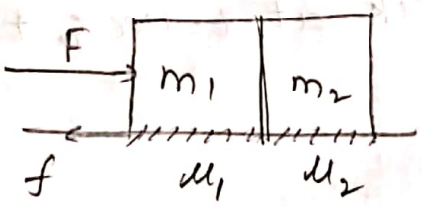


$$f_{2l} = \mu_2 m_2 g \qquad f_{1l} = \mu_1 m_1 g$$

$$F > f_{1l}$$

$$\boxed{F - f_{1l} - kx = m_1 a} \rightarrow (1)$$

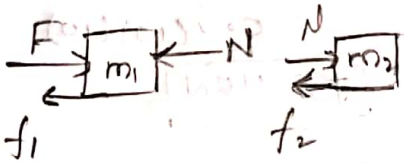
Contact bodies:



$$f_{1l} = \mu_1 m_1 g$$

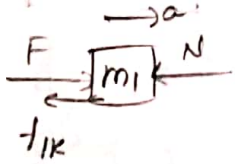
$$f_{2l} = \mu_2 m_2 g$$

* If $F < (f_{1l} + f_{2l})$, system is at rest.



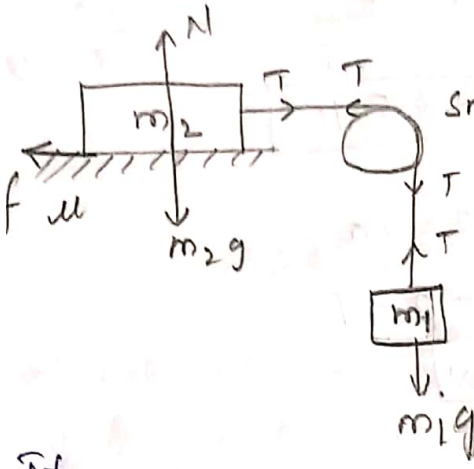
* If $F > (f_{1l} + f_{2l})$, system moves with acceleration 'a'.

$$a = \frac{F - (f_{1k} + f_{2k})}{m_1 + m_2}$$



$$\boxed{F - f_{1k} - N = m_1 a}$$

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Block m_2 is at rest,

$T < f_L$

$T < \mu m_2 g$

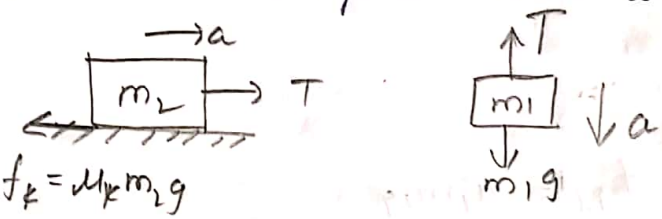
again $T = m_1 g$.

$m_1 g < \mu m_2 g$

$\boxed{m_1 < \mu m_2}$

Here $\boxed{f = T = m_1 g}$

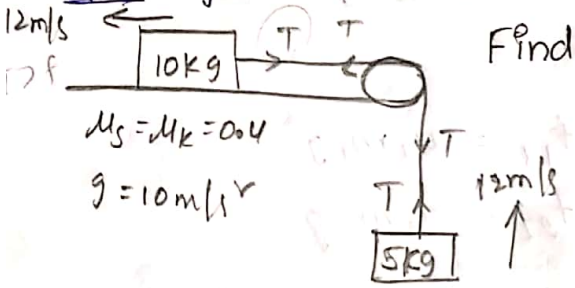
If $m_1 > \mu m_2$, system moves with acceleration 'a'.



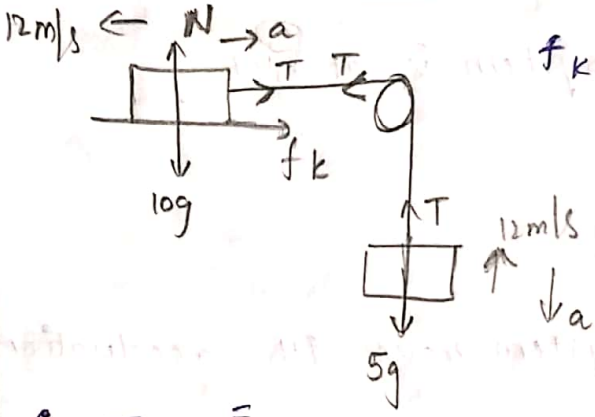
$T - \mu_k m_2 g = m_2 a$ —①

$m_1 g - T = m_1 a$ —②

solve for T & a.



Find velocity & position of 10kg block at t=1sec & t=4sec.



$f_k = \mu_k N = \mu_k 10g$

$= 0.4(10)(10)$

$\boxed{f_k = 40N}$

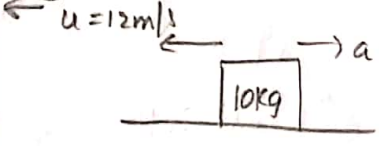
$T + f_k = 10a$

$5g - T = 5a$

$5g + f_k = 15a$

$90 = 15a \Rightarrow \boxed{a = 6 m/s^2}$

& $T + 40 = 60 \Rightarrow \boxed{T = 20N}$

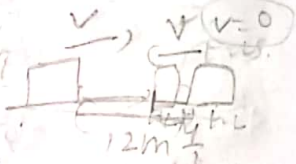


$v = u + at$

$= 12 - 6t$

$v = 0$ when $t = 2 sec$

at $t = 1$ sec $v = 12 - 6(1) = 6 m/s$ (⟵)



& displacement of block in 1sec $\Rightarrow s = ut + \frac{1}{2}at^2$

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$S = 12(1) - \frac{1}{2}(6)(1)^2$

$= 12 - 3$

$\boxed{S = 9m}$ ∴ It's position is at 9m from starting point to left.

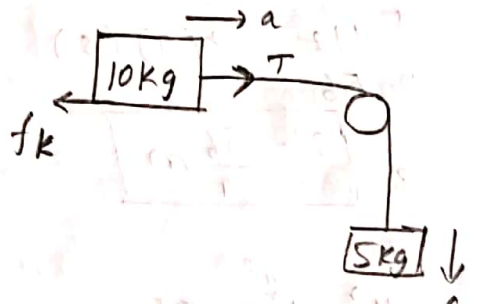
* Total distance travelled in $t = 2$ sec,

$S = (12)(2) - \frac{1}{2}(6)(4)$

$= 24 - 12$

$\boxed{S = 12m}$, to left.

* For $t > 2$ sec,



$T - f_k = 10a$

$T - 40 = 10a$ —①

$5g - T = 5a$ —②

$5g - 40 = 15a$

$10 = 15a$

$\boxed{a = 2/3\ m/s^2}$

& $T = (4 + \frac{2}{3})10 = \frac{140}{3}$ N [∵ from ①]

* Here at $t = 2$ sec,

$v = u + at$

$v = 0 + (\frac{2}{3})(2)$

$\boxed{v = 4/3\ m/s}$ (⟶)

& $S = ut + \frac{1}{2}at^2$

$= \frac{1}{2}(\frac{2}{3})(4)$

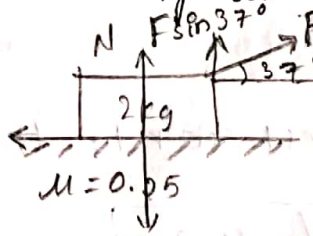
$= 4/3$ m

∴ It's position at $t = 4$ sec is

$= 12 - \frac{4}{3} = \frac{32}{3}$ m to left from starting point.

→ After the block has moved a distance 8m to the right, the direction of horizontal component of force 'F' is reversed. Find the velocity with which block arrived at the starting point.

*



$F\cos 37° = \frac{20 \times 4}{5} = 16$

$N + F\sin 37° = 2g$ ⇒ $\boxed{N = 8N}$

$2a + F\cos 37° = f_k$

$2a + 20 \times \frac{4}{5} = 8$

$a + 8 = 1$

$a = -7\ m/s^2$

⇒ $a = 7\ m/s^2$ (⟵)

$f_k = 0.25(20 - 20 \times \frac{3}{5})$

$= 0.25(20 - 12)$

$= 0.25(8)$

$= 2.00 = 2N$

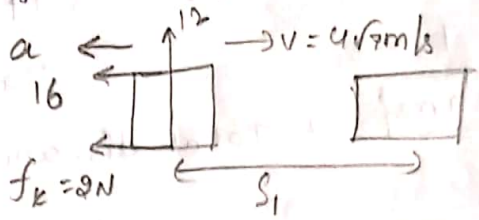
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$v^2 - u^2 = 2as$

$v^2 = 2(7)(8)$

$\boxed{v = 4\sqrt{7}\ m/s}$

*After 8m,



$a = \frac{16+2}{2} = 9\ m/s^2$

$v^2 - u^2 = 2as$

$-(4\sqrt{7})^2 = 2(-9)(s)$

$-112 = -2(9)(s)$

56

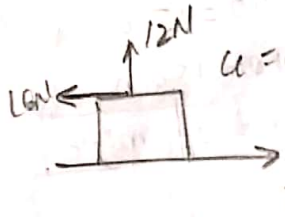
$\boxed{s = \frac{56}{9}\ m}$

* Total distance travelled before coming to rest $= 8 + \frac{56}{9}$

$= \frac{72+56}{9}$

$= \frac{128}{9}\ m$

*



$a = \frac{16-2}{2} = \frac{14}{2} = 7\ m/s^2$

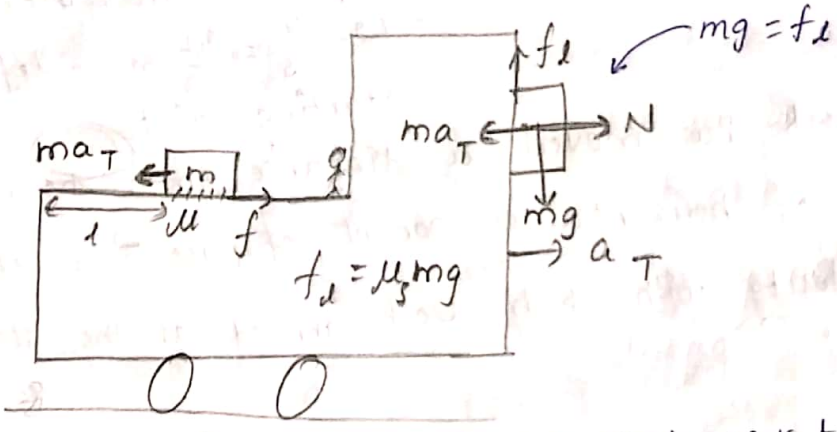
$v^2 - u^2 = 2as$

$v^2 = 2(7)\left(\frac{128}{9}\right)$

$v = \frac{24}{3}\sqrt{7}$

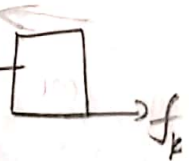
$\boxed{v = \frac{16\sqrt{7}}{3}\ m/s}$

Block on accelerating truck:



* $ma_T < f_l$, block is at rest w.r.to truck & friction is static. $\Rightarrow \boxed{f = ma_T}$

* $ma_T = f_l$; block is ready to slide on truck.

* $ma_T = f_l$  If same force is continued to act on block even after motion has started.

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then, $a_{bT} = \frac{f_s - f_k}{m} = \frac{(\mu_s - \mu_k) mg}{m}$

$$\boxed{a_{bT} = (\mu_s - \mu_k) g}$$

* If $ma_T > f_s$; block has acceleration on truck

$$\therefore a_{bT} = \frac{ma_T - f_k}{m} = \frac{ma_T - \mu_k mg}{m}$$

$$\boxed{a_{bT} = a_T - \mu_k g} \quad (\leftarrow)$$

* time taken by block to seperate from the truck, $s_{rel} = u_{rel}^{\circ} t + \frac{1}{2} a_{rel} t^2$

$$l = \frac{1}{2} (a_T - \mu_k g) t^2$$

$$\boxed{t = \sqrt{\frac{2l}{a_T - \mu_k g}}}$$

* speed of block while seperating the truck,

$$v_{bT} = u_{bT}^{\circ} + a_{bT} t$$

$$v_{bT} = (a_T - \mu_k g) \sqrt{\frac{2l}{a_T - \mu_k g}}$$

$$\boxed{v_{bT} = \sqrt{2l(a_T - \mu_k g)}}$$

* Velocity of block,

$$\vec{v_b} = \vec{v_{bT}} + \vec{v_T}$$

$$\vec{v_b} = v_{bT} (-\uparrow) + v_T (\uparrow)$$

$$\boxed{\vec{v_b} = (v_T - v_{bT}) (\uparrow)}$$

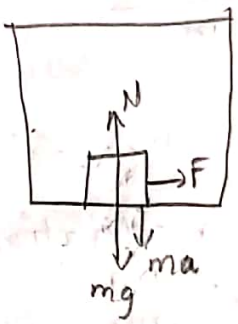
* distance travelled by block,

$$\vec{s_b} = \vec{s_{bT}} + \vec{s_T}$$

$$\vec{s_b} = l (-\uparrow) + s_T (\uparrow) \Rightarrow \boxed{\vec{s_b} = (s_T - l) (\uparrow)}$$

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for lift:

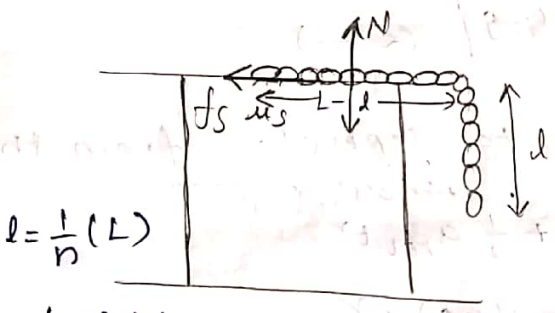


$N = (m)(g+a)$

$F = \mu_s N = \mu(m)(g+a)$

$\boxed{F = \mu m(g+a)}$

Chain on the table :



if the nth part of the chain hanging from the edge of table.

It is in limiting equilibrium.

$L \to M$

$l \to m \Rightarrow m = \frac{M}{L} \times l$

$\boxed{m' = \frac{M}{L}(L-l)}$

$\omega = f_s$

$\frac{M}{L} lg = \mu_s N$

$\frac{M}{L} lg = \mu_s \left(\frac{M}{L}(L-l)g\right)$

$\boxed{l = \mu_s(L-l)}$

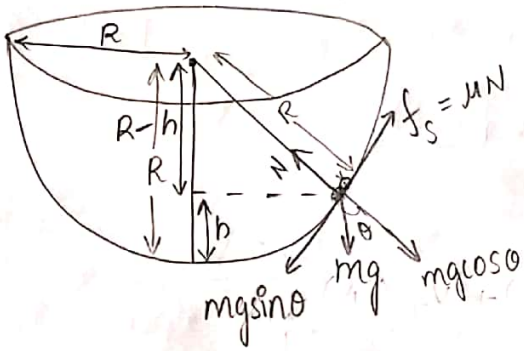
$\mu_s = \frac{l}{L-l} = \frac{1}{\frac{L}{l}-1}$

$\frac{1}{\mu_s} = \frac{L}{l} - 1 \Rightarrow \frac{L}{l} = \frac{1}{\mu_s} + 1$

$\boxed{\mu_s = \frac{1}{n-1}}$

$\boxed{\frac{l}{L} = \frac{\mu_s}{\mu_s + 1}}$

Insect crawling in a bowl :-



In equilibrium,

$N = mg\cos\theta$ &

$f_s = mg\sin\theta = \mu_s N$

$\mu mg\cos\theta = mg\sin\theta$

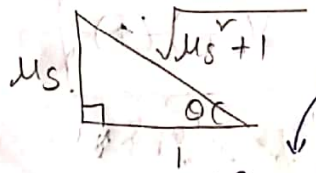
$\boxed{\mu_s = \tan\theta}$

$\cos\theta = \frac{R-h}{R}$

$\cos\theta = 1 - \frac{h}{R}$

$\frac{h}{R} = 1 - \cos\theta$

$\boxed{h = R(1-\cos\theta)}$



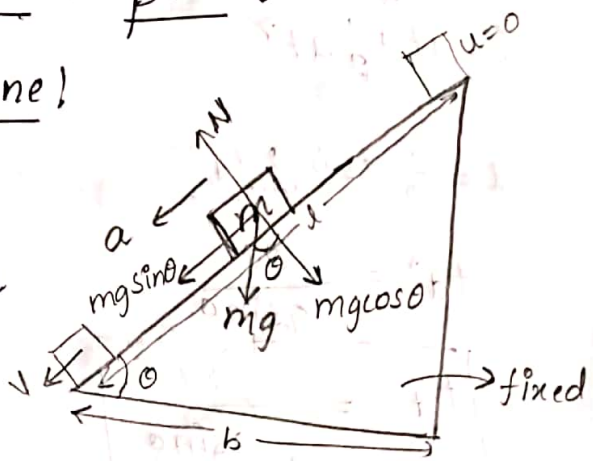
$\boxed{h = R\left(1 - \frac{1}{\sqrt{1+\mu_s^2}}\right)}$

max height upto which insect can crawl in the bowl.

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Block placed on inclined plane:-

① Smooth Inclined Plane:



* $a = \frac{mg\sin\theta}{m} = g\sin\theta$

* velocity of block on reaching bottom,

$v^2 - u^2 = 2as$

$v^2 - 0^2 = 2g\sin\theta\, l$

$$\boxed{v = \sqrt{2gl\sin\theta}}$$

From figure,

~~t=tmin when~~

$\cos\theta = \frac{b}{l} \Rightarrow l = \frac{b}{\cos\theta}$

* time taken to reach bottom,

$l = \frac{1}{2}at^2$

$t^2 = \frac{2l}{g\sin\theta}$

$$\boxed{t = \sqrt{\frac{2l}{g\sin\theta}}}$$

$$\boxed{t = \sqrt{\frac{2b}{g\sin\theta\cos\theta}}} \Rightarrow \boxed{\sqrt{\frac{4b}{g\sin 2\theta}} = t}$$

$v = \sqrt{2gb\tan\theta}$

*



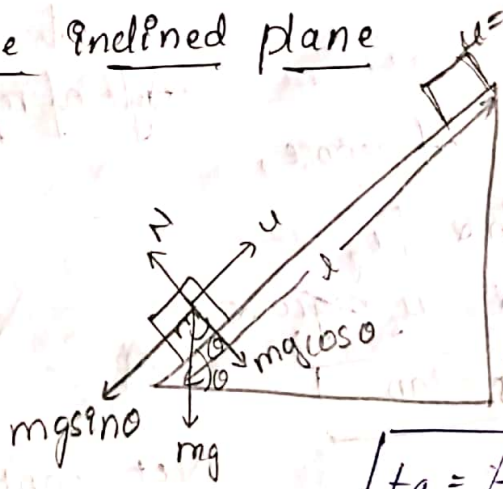
For a given base length,

$t = t_{min}$ when $\sin 2\theta = 1$

$2\theta = 90°$

$\theta = 45°$

Projected up the inclined plane



* $a = -g\sin\theta$

* $v = u + at$

$0 = u - g\sin\theta\, t$

$$\boxed{t_a = \frac{u}{g\sin\theta}}$$

↳ time of ascent.

$v^2 - u^2 = 2as$

$$\boxed{u = \sqrt{2gl\sin\theta}}$$

$$t_a = \sqrt{\frac{2gl\sin\theta}{g^2\sin^2\theta}}$$

$$\boxed{t_a = \sqrt{\frac{2l}{g\sin\theta}}}$$

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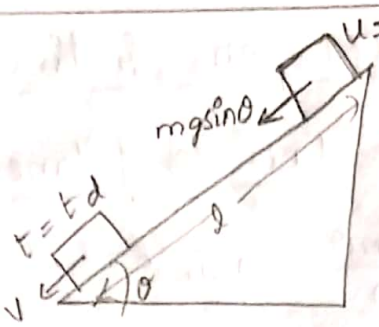
$S = \cancel{ut}^{0} + \frac{1}{2}at^2$

$l = \frac{1}{2} g\sin\theta\, t_d^2$

$t_d^2 = \frac{2l}{g\sin\theta}$

$$\boxed{t_d = \sqrt{\frac{2l}{g\sin\theta}}}$$

clearly $t_a = t_d$.



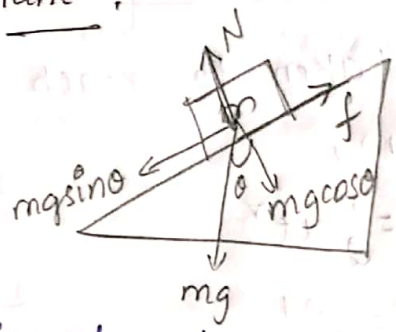
$$\boxed{a = g\sin\theta}$$

* velocity on reaching the bottom, $v^2 - \cancel{u^2}^{0} = 2as$

$$\boxed{v = \sqrt{2.lg\sin\theta}}$$

Clearly $\boxed{v = u}$ *

(2) Rough Inclined Plane :



Here $N = mg\cos\theta$;

$f_l = \mu_s (mg\cos\theta)$

** $mg\sin\theta < f_l$; Block is at rest.

friction → static.

$f = mg\sin\theta$



$R = \sqrt{N^2 + f^2}$

$R = \sqrt{(mg\cos\theta)^2 + (mg\sin\theta)^2}$

$$\boxed{R = mg}$$

* As $\theta\uparrow \Rightarrow \sin\theta\uparrow \Rightarrow mg\sin\theta\uparrow$ & $f\uparrow$

when $\theta = \alpha$, then $mg\sin\alpha = f_l$

↓

Angle of repose → Block begins to slide.

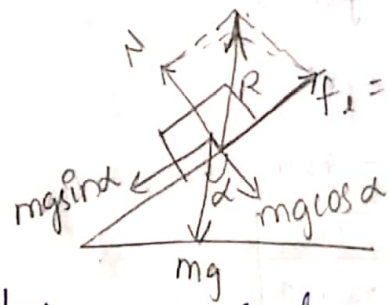
* At angle of repose,

$mg\sin\alpha = f_l$

$\cancel{mg}\sin\alpha = \mu_s \cancel{mg}\cos\alpha$

$$\boxed{\mu_s = \tan\alpha}$$

$\alpha = \phi$



Net contact force, $R = \sqrt{N^2 + f_l^2}$

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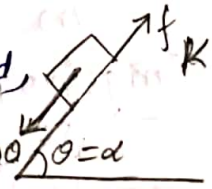
$$R = \sqrt{N^2 + (\mu_s N)^2} = N\sqrt{1+\mu_s^2}$$

$$\boxed{R = mg\cos\theta\sqrt{1+\mu_s^2}}$$

* If $\theta < \alpha$; Block is at rest (if it come to rest)

* If $\theta = \alpha$; Block is ready to slide.

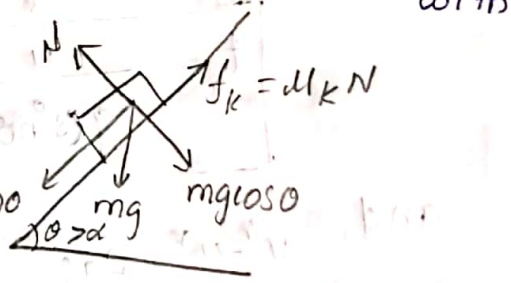
* even after motion has started, same force ($f_1 = mg\sin\theta$) is acting on it. Then accelesation of block:



$$a = \frac{f_1 - f_k}{m} = \frac{mg\sin\theta - \mu_s N - \mu_k N}{m}$$

$$\boxed{a = (\mu_s - \mu_k) g\cos\theta}$$

* If $\theta > \alpha \Rightarrow mg\sin\theta > f_1$; Block sides down with acceleration 'a'.

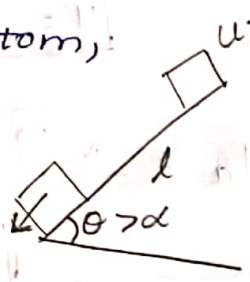


$$a = \frac{mg\sin\theta - \mu_k N}{m}$$

$$a = \frac{mg\sin\theta - \mu_k mg\cos\theta}{m}$$

** $\boxed{a = g(\sin\theta - \mu_k\cos\theta)}$

* Velocity on reaching the bottom,



$$v^2 - u^2 = 2as$$

$$\boxed{v = \sqrt{2gl(\sin\theta - \mu_k\cos\theta)}}$$

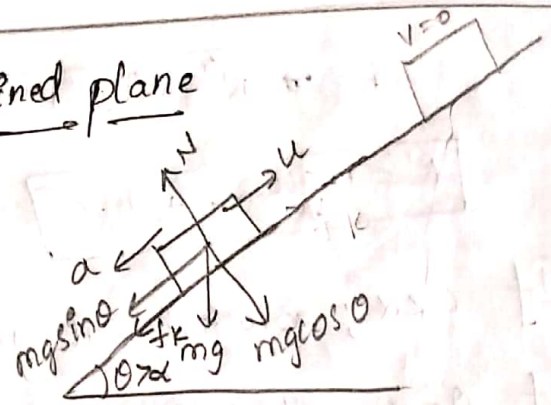
time ~~taken~~ taken to reach the bottom;

$$s = ut + \frac{1}{2}at^2 \Rightarrow l = \frac{1}{2}g(\sin\theta - \mu_k\cos\theta)t^2$$

$$\boxed{t = \sqrt{\frac{2l}{g(\sin\theta - \mu_k\cos\theta)}}}$$

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Projected up the inclined plane



$N = mg\cos\theta$

$f_k = \mu_k N = mg\mu_k\cos\theta$

* $a = -\dfrac{(mg\sin\theta + f_k)}{m}$

$a = -\dfrac{(mg\sin\theta + \mu_k mg\cos\theta)}{m}$

$$\boxed{a = -g(\sin\theta + \mu_k\cos\theta)}$$

Clearly decelerating is more than acceleration.

* Time of ascent,

$v = u + at$

$0 = u - g(\sin\theta + \mu_k\cos\theta)\, t_a$

$$\boxed{t_a = \frac{u}{g(\sin\theta + \mu_k\cos\theta)}}$$

and $v^2 - u^2 = 2as$

$-u^2 = 2(-g(\sin\theta + \mu_k\cos\theta))\, l$

$$\boxed{u = \sqrt{2gl(\sin\theta + \mu_k\cos\theta)}}$$

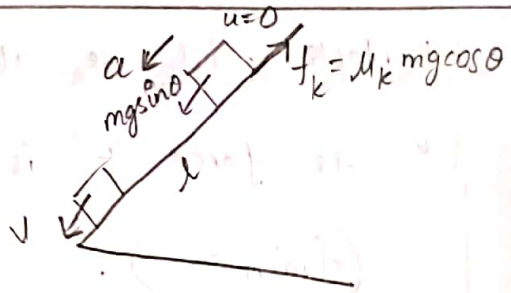
$$t_a = \frac{\sqrt{2gl(\sin\theta + \mu_k\cos\theta)}}{g(\sin\theta + \mu_k\cos\theta)}$$

$$\boxed{t_a = \frac{\sqrt{2gl}}{\sqrt{g(\sin\theta + \mu_k\cos\theta)}}}$$

* Time of descent,

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$$a = g(\sin\theta - \mu_k \cos\theta)$$



$$s = ut + \frac{1}{2}at^2$$

$$l = \frac{1}{2}g(\sin\theta - \mu_k\cos\theta)\,t_d^2$$

$$\boxed{t_d = \sqrt{\frac{2l}{g(\sin\theta - \mu_k\cos\theta)}}}$$

Clearly $\boxed{t_a < t_d}$

* Velocity on reaching bottom;

$$v^2 - u^2 = 2as$$

$$v^2 - 0^2 = 2g(\sin\theta - \mu_k\cos\theta)\,l$$

$$\boxed{v = \sqrt{2gl(\sin\theta - \mu_k\cos\theta)}}$$

Clearly $\boxed{u > v}$

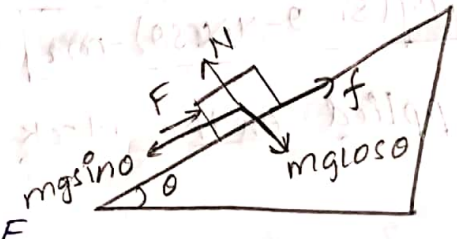
Impending Motion:

* Force is applied parallel to inclined plane:

(a) Impending motion down the plane:

$$mg\sin\theta = F + f_l$$

$$\Rightarrow mg\sin\theta = F + \mu_s N$$



$N = mg\cos\theta$

$$mg\sin\theta - \mu_s mg\cos\theta = F$$

$$mg(\sin\theta - \mu_s\cos\theta) = F$$

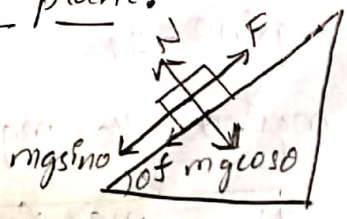
$$\boxed{F_{min} = mg(\sin\theta - \mu_s\cos\theta)}$$

min force required to keep the block at rest on inclined plane.

(b) Impending motion up the plane:

$$F = mg\sin\theta + f_l$$

$$= mg\sin\theta + \mu_s(mg\cos\theta)$$



$$\boxed{F_{max} = mg(\sin\theta + \mu_s\cos\theta)}$$

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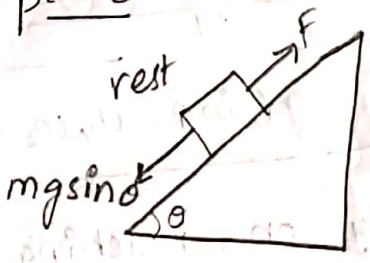
* Block will be at rest for, $F_{min} \leq F \leq F_{max}$ (b)

* As force 'F' is increases, when $F = mg\sin\theta$; [for (a)]

$f_{min} = 0$.

* for smooth inclined plane.

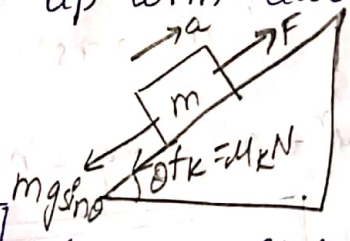
$F = mg\sin\theta$



* If $F > F_{max}$; block moves up with acceleration 'a'.

$F - mg\sin\theta - \mu_K N = ma$

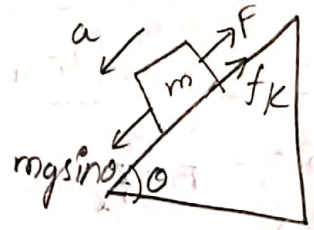
$F = mg(\sin\theta + \mu_K \cos\theta) + ma$ → force required to pull the block up the plane with acceleration a.



* If $F < F_{max}$; block moves down with acceleration a.

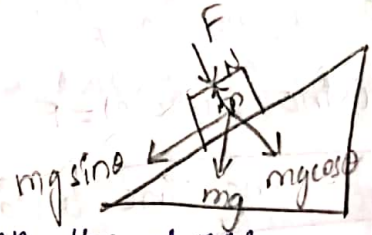
$mg\sin\theta - F - f_K = ma$

$F = mg(\sin\theta - \mu_K \cos\theta) - ma$



Force applied on the block normal to the inclined plane ;

$N = mg\cos\theta + F$ —①



*for impending motion down the plane:

$mg\sin\theta = f_s$

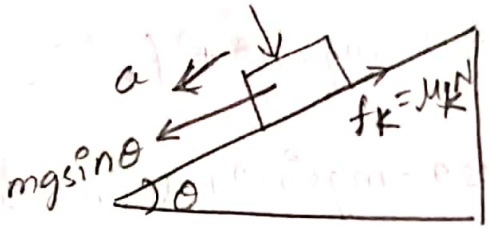
$mg\sin\theta = \mu_s (mg\cos\theta + F)$

$F_{min} = \dfrac{mg(\sin\theta - \mu_s\cos\theta)}{\mu_s}$

min force to keep the block at rest.

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* If $F < F_{min}$; block slides down with acceleration a.



$$mg\sin\theta - f_k = ma$$

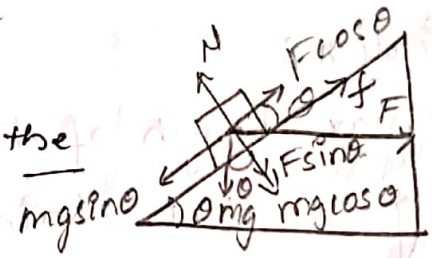
$$mg\sin\theta - \mu_k(mg\cos\theta + F) = ma$$

$$mg(\sin\theta - \mu_k\cos\theta) - ma = \mu_k F$$

$$\boxed{F = \frac{mg(\sin\theta - \mu_k\cos\theta) - ma}{\mu_k}}$$

* Force applied on the block in horizontal direction :

(a) Impending motion down the inclined plane :



$$N = mg\cos\theta + F\sin\theta$$

$$mg\sin\theta = F\cos\theta + f$$

$$mg\sin\theta = F\cos\theta + \mu_s(mg\cos\theta + F\sin\theta)$$

$$mg\sin\theta - \mu_s mg\cos\theta = F(\cos\theta + \mu_s\sin\theta)$$

$$\boxed{F_{min} = \frac{mg(\sin\theta - \mu_s\cos\theta)}{\cos\theta + \mu_s\sin\theta}}$$

→ min force required to keep block at rest.

* As F ↑ ⇒ friction ↓ :

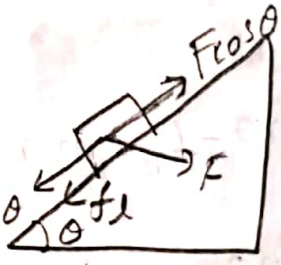
when $F\cos\theta = mg\sin\theta$

$$\boxed{F = mg\tan\theta} \Rightarrow \text{friction} = 0$$

* Still as F ↑ ⇒ friction changes its direction & also ↑.

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(b) Impending motion up the inclined plane:



$F\cos\theta = mg\sin\theta + f_s$

$F\cos\theta = mg\sin\theta + \mu_s(mg\cos\theta + F\sin\theta)$

$F\cos\theta = mg(\sin\theta + \mu_s\cos\theta) + \mu_s F\sin\theta$

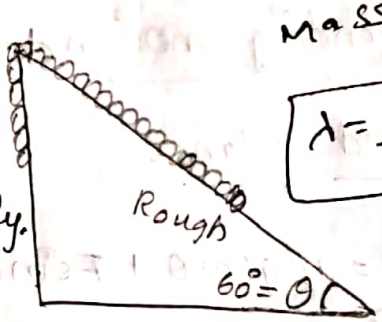
$F(\cos\theta - \mu_s\sin\theta) = mg(\sin\theta + \mu_s\cos\theta)$

$$F_{max} = \frac{mg(\sin\theta + \mu_s\cos\theta)}{\cos\theta - \mu_s\sin\theta}$$

* Block is at rest for $F_{min} \le F \le F_{max}$

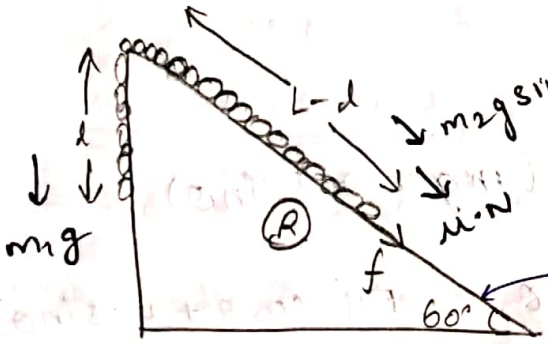
Application:

Find max length of chain hanging so that chain has a tendency to fall vertically.



Mass per unit length

$$\lambda = \frac{M}{L}$$



L – total length of chain

λ – mass per unit length = $\frac{M}{L}$

rough inclined plane.

$\lambda l g = \lambda(L-l)g\sin 60° + \mu\lambda(L-l)g\cos 60°$

$l = (L-l)\frac{\sqrt{3}}{2} + \frac{\mu}{2}(L-l)$

$l = \frac{(L-l)}{2}(\sqrt{3}+\mu)$

$l\left(\frac{2+\sqrt{3}+\mu}{\sqrt{3}+\mu}\right) = L$

$$l = \frac{(\sqrt{3}+\mu)(L-l)}{2}$$

$$l = L\left(\frac{\sqrt{3}+\mu}{2+\sqrt{3}+\mu}\right)$$

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* fractional length of chain hanging :-

$$\frac{l}{L} = \frac{\sqrt{3}+\mu}{2+\sqrt{3}+\mu}$$

% of length of chain hanging

$$\frac{l}{L} \times 100 = \left(\frac{\sqrt{3}+\mu}{2+\sqrt{3}+\mu}\right) \times 100 \%$$

→ Find the length of chain hanging so that chain has tendency to slide down the plane.

$$\lambda l g + f_l = \lambda (L-l) g \sin 60^\circ$$

$$\lambda l g + \mu \lambda (L-l) g \cos 60^\circ = \lambda (L-l) g \sin 60^\circ$$

$$l + \mu (L-l) \times \frac{1}{2} = (L-l) \frac{\sqrt{3}}{2}$$

$$l = (L-l)\left(\frac{\sqrt{3}-\mu}{2}\right)$$

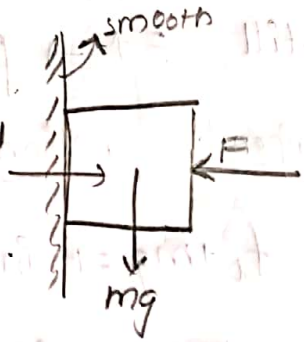
$$l\left(\frac{2+\sqrt{3}-\mu}{\sqrt{3}-\mu}\right) = L$$

$$\boxed{l = \left(\frac{\sqrt{3}-\mu}{2+\sqrt{3}-\mu}\right) L}$$

Block in contact with Vertical surface:

* Here $N = F$.
* Block will not be in equilibrium for any large value of 'F' may be.

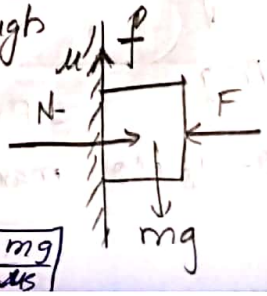
& $a = \frac{mg}{m} = g (\downarrow)$



* Block will be at rest. i.e,

$mg \le f_l$

$mg \le \mu_s N \quad mg \le \mu_s F \quad \boxed{F \ge \frac{mg}{\mu_s}}$



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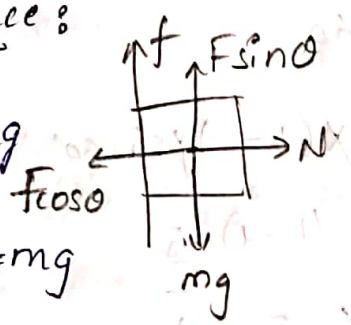
* If $mg > f_l$ or $F < \frac{mg}{\mu_s}$; Block slides down with acceleration a.

$$a = \frac{mg - f_k}{m}$$

$$a = \frac{mg - \mu_k N}{m}$$

$$\boxed{a = \frac{mg - \mu_k F}{m}}$$

Impending motion down the surface:

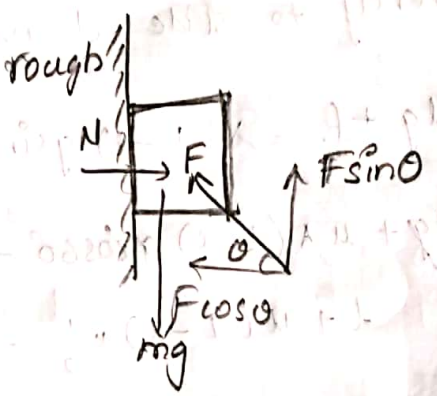


$f_l + F\sin\theta = mg$

$\mu_s N + F\sin\theta = mg$

$\mu_s(F\cos\theta) + F\sin\theta = mg$

$$\boxed{F_{min} = \frac{mg}{\mu_s\cos\theta + \sin\theta}}$$

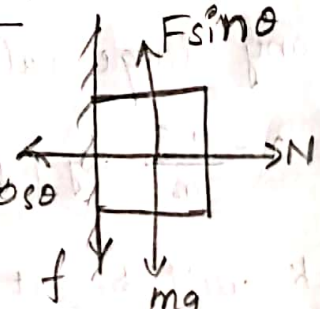


* As $F\uparrow \Rightarrow f\downarrow$ in magnitude only.

when $\boxed{F\sin\theta = mg}$ then $\boxed{f = 0}$

* still $F\uparrow \Rightarrow f\uparrow$ in opp direction.

Impending motion up the surface:



$f_l + mg = F\sin\theta$

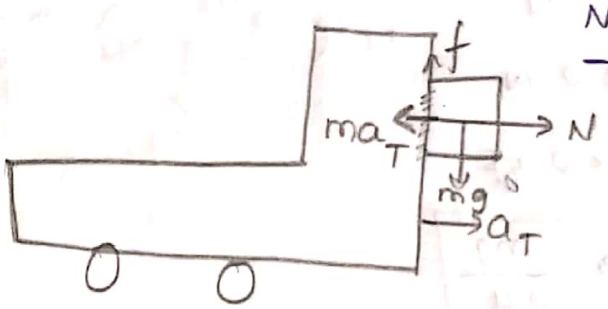
$mg = F\sin\theta - \mu_s(F\cos\theta)$

$$\boxed{\frac{mg}{\sin\theta - \mu_s\cos\theta} = F_{max}}$$

∴ Block will be at rest for $F_{min} \le F \le F_{max}$.

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Accelerating truck:-



$N = ma_T$

* Block will be at rest w.r.to truck,

$$mg \leq f_l$$

$$f_l = \mu_s N \Rightarrow mg \leq \mu_s N$$

$$mg \leq \mu_s (ma_T)$$

$$\boxed{a_T \geq g/\mu_s}$$

min acceleration to keep block at rest $= g/\mu_s$.

* If $mg > f_l$; Block slides down with an acceleration "a_{bT}".

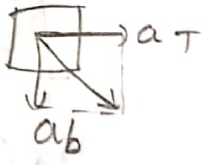
$$a_{bT} = \frac{mg - f_k}{m}$$

$$a_{bT} = \frac{mg - \mu_k ma_T}{m}$$

$$\boxed{a_{bT} = g - \mu_k a_T}$$

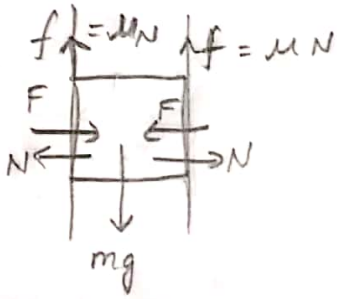
net acceleration of block:

$$\vec{a_b} = \vec{a_{bT}} + \vec{a_T}$$



$$\boxed{a_b = \sqrt{a_{bT}^2 + a_T^2}}$$

Book pressing by two hands:



In limiting equilibrium,

$mg = 2\mu N$

$mg = 2\mu F$ $\quad \boxed{F = \frac{mg}{2\mu}}$

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* $N_1 = F_1$

$N_2 = F_2$

$N_1 - N_2 = 0$

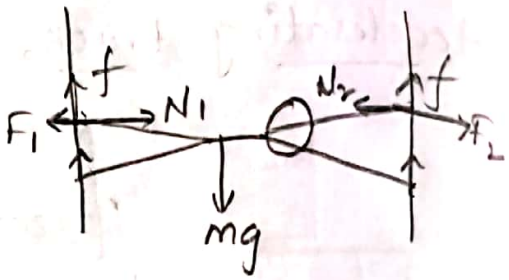
$\boxed{N_1 = N_2}$

$mg \leq 2f_1$

$mg \leq 2\mu N$

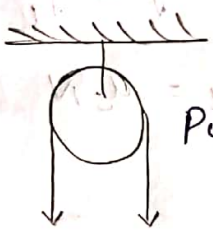
$mg \leq 2\mu_s F$

$\boxed{F \geq \frac{mg}{2\mu}}$



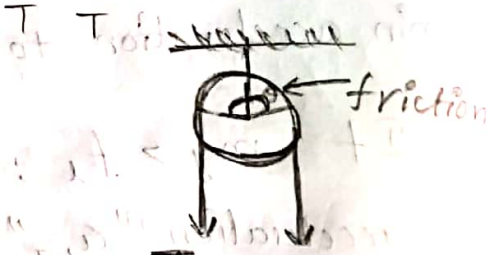
Friction b/w Pulley and string :-

* String slips on pulley and pulley is at rest.



Pulley is smooth.

* There is no slipping b/w string & pulley.

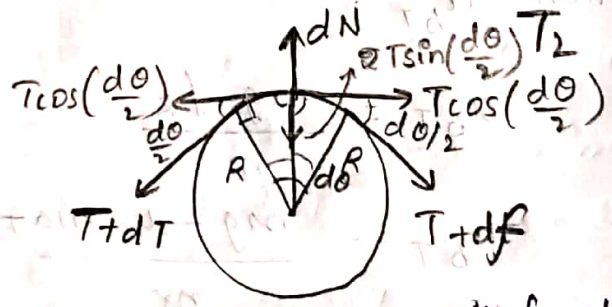


* Suppose $T_2 > T_1$

$dN = 2T\sin\left(\frac{d\theta}{2}\right)$

$dN = 2T\left(\frac{d\theta}{2}\right)$

$\boxed{dN = T\,d\theta}$



* for limiting friction,

$df = \mu dN$

$dT = \mu dN$

$dT = \mu T\,d\theta$

$\int_{T_1}^{T_2} \frac{dT}{T} = \int_0^{\theta} \mu \cdot d\theta$

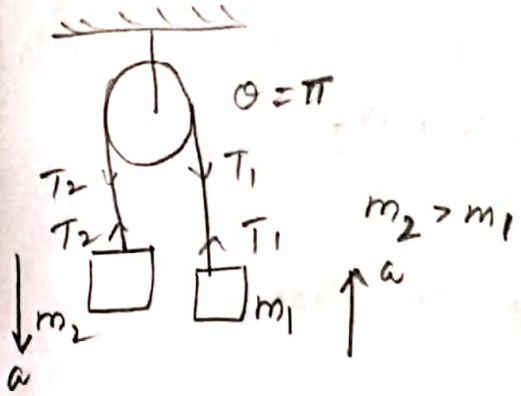
$\ln(T_2/T_1) = \mu\theta$

$\frac{T_2}{T_1} = e^{\mu\theta}$

$\boxed{T_\theta = T_1 e^{\mu\theta}}$

$\theta \rightarrow$ angle of contact.

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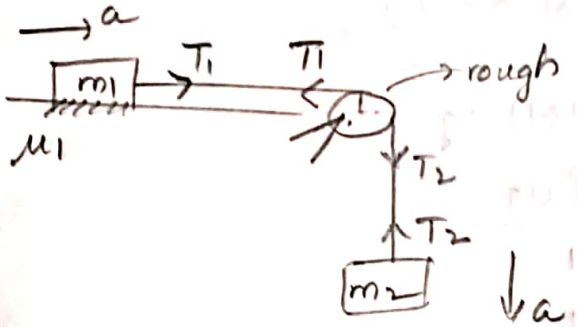
$m_2 g - T_2 = m_2 a$ — ①

$T_1 - m_1 g = m_1 a$ — ②

$T_2 = m_2 (g - a)$

$T_1 = m_1 (g + a)$

$T_2 = T_1 e^{\mu\theta}$ — ③



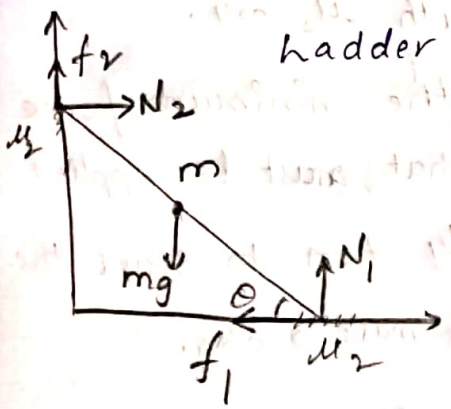
$m_2 g - T_2 = m_2 a$ — ①

$T_1 - \mu_1 m_1 g = m_1 a$ — ②

$T_2 = T_1 e^{\mu\theta}$ — ③

Equilibrium: $\Sigma F_{net} = 0$

$\Sigma F_V = 0$ & $\Sigma F_H = 0$



ladder is in limiting equilibrium.

Here $f_2 = \mu_2 N_2$, $f_1 = \mu_1 N_1$.

$\Sigma F_H = 0$ $\quad$ $\Sigma F_V = 0$

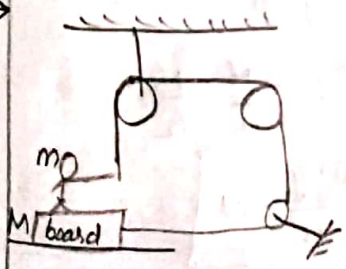
$N_2 = F_1$ $\quad$ $N_1 + f_2 = mg$

$N_2 = \mu_1 N_1$ — ① $\quad$ $N_1 + \mu_2 N_2 = mg$ — ②

② ⇒ $N_1 + \mu_1 \mu_2 N_1 = mg$

$$N_1 = \frac{mg}{1 + \mu_1 \mu_2}$$

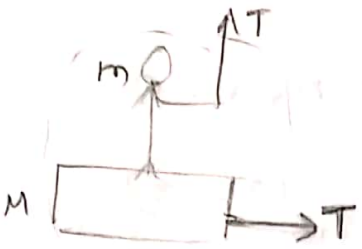
① ⇒ $$N_2 = \frac{\mu_1 mg}{1 + \mu_1 \mu_2}$$



Coefficient of friction blw board & floor is μ. The maximum force that the man can exert on the rope so that the board does not slip on floor.

Take man + board as system.

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* $\Sigma F_y = 0$

$$N + T = (m+M)g \quad \text{——①}$$

Since board doesn't slip on floor,

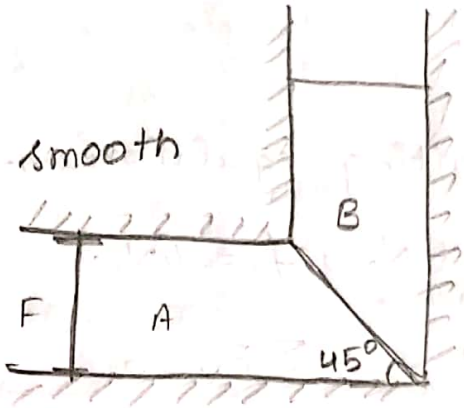
$$T \leq \mu N$$

$$T \leq \mu((m+M)g - T)$$

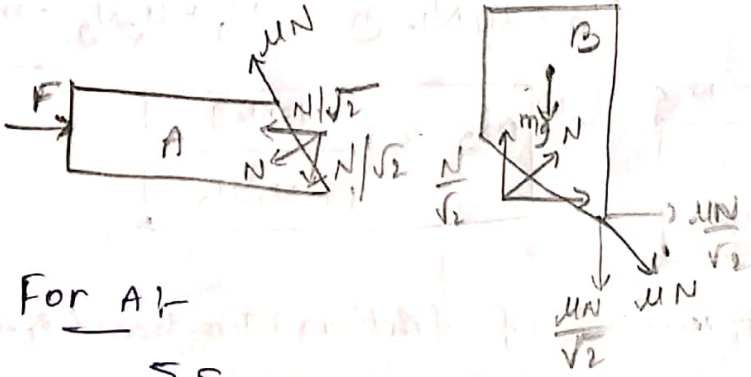
$$T + \mu T \leq (m+M)\mu g$$

$$T \leq \frac{(m+M)\mu g}{\mu+1}$$

* $$\boxed{T_{max} = \frac{(m+M)\mu g}{\mu+1}}$$



Friction exists at interface of A & B with $\mu_s = 0.4$. what is the minimum force F (in N) that must be applied horizontally to A to start the motion of latch B if it has mass 0.6 kg.



$\Sigma F_x = 0, \Sigma F_y = 0$

For A:-

$\Sigma F_x = 0$

$$F = \frac{N}{\sqrt{2}} + \frac{\mu N}{\sqrt{2}}$$

$$= \frac{N}{\sqrt{2}}(1+\mu) = \frac{mg(1+\mu)}{1-\mu}$$

For B:-

$\Sigma F_y = 0$

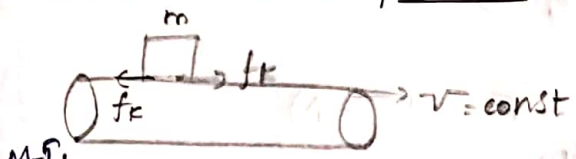
$$\Rightarrow mg + \frac{\mu N}{\sqrt{2}} = \frac{N}{\sqrt{2}}$$

$$\boxed{\frac{\sqrt{2}mg}{1-\mu} = N}$$

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$$F = \frac{0.6 \times 10 (1.4)}{0.6} = 14N.$$

→ Conveyer belt problems:





M-I:

* $a_{block} = \frac{f_k}{m} = \frac{\mu mg}{m} = \mu g$

$a_{belt} = 0$

* Time taken by the block to come to rest on belt, $v = u + at$

$$v = \mu g t$$

$$\boxed{t = \frac{v}{\mu g}} \quad **$$

* In this time,

distance travelled by belt, $S_b = vt$

$$\boxed{S_{belt} = \frac{v^2}{\mu g}}$$

$s = \frac{v^2}{\mu g} \times \mu g \times \frac{1}{2}$

$s = \frac{v^2}{2\mu g}$,

& distance travelled by block,

$$S_{block} = ut + \frac{1}{2} a t^2$$

$$S_{block} = \frac{1}{2} (\mu g) \frac{v^2}{\mu^2 g^2}$$

$$\boxed{S_{block} = \frac{1}{2} \frac{v^2}{\mu g}} \quad \text{(w.r.to ground)}$$

* displacement of block w.r.t belt, when it comes to rest on it.

B → Block
b → belt

$$\vec{S}_{B|b} = \vec{S}_B - \vec{S}_b$$

$$= \frac{v^2}{2\mu g} (\uparrow) - \frac{v^2}{\mu g} (\uparrow) = \frac{v^2}{2\mu g} (-\uparrow)$$

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$$\boxed{\bar{s}_{B/b} = \frac{V^2}{2\mu g}(\leftarrow \uparrow)}$$

M-II (relative motion concept)

$$\bar{u}_{rel} = \bar{u}_{B/b} = \bar{u}_B - \bar{u}_b = -\bar{V} = V(\leftarrow\uparrow)$$

$$\bar{a}_{rel} = \bar{a}_{B/b} = \bar{a}_B - \bar{a}_b = \mu g(\uparrow)$$

* time for block is at rest w.r.t belt, $[v_{rel}=0]$

$$\bar{v}_{rel} = \bar{u}_{rel} + \bar{a}_{rel}\, t$$

$$0 = -V\uparrow + \mu g \uparrow (t)$$

$$\boxed{\frac{V}{\mu g} = t}\ *$$

* displacement of block w.r.t. belt in time $t = \frac{V}{\mu g}$.

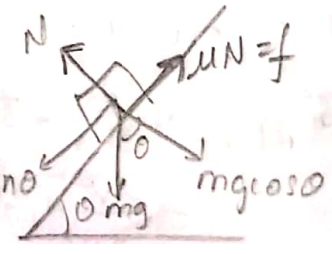
$$\bar{s}_{rel} = \bar{u}_{rel}\, t + \frac{1}{2}\bar{a}_{rel}\, t^2$$

$$= -V(\uparrow)\frac{V}{\mu g} + \frac{1}{2}\mu g(\uparrow)\left(\frac{V^2}{\mu^2 g^2}\right)$$

$$= \frac{-V^2}{\mu g}\uparrow + \frac{1}{2}\frac{V^2}{\mu g}\uparrow$$

$$\boxed{\bar{s}_{rel} = -\frac{V^2}{2\mu g}(\leftarrow\uparrow)}$$

Block is kept symmetrically in the channel & μ is coefficient of friction b/w block & channel, find acceleration of block.



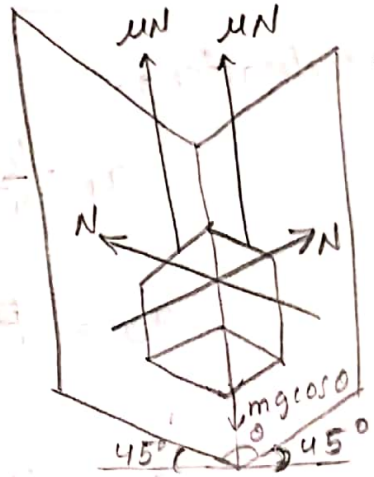
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$\sqrt{2}N = mg\cos\theta \rightarrow (1)$

$mg\sin\theta - 2\mu N = ma$

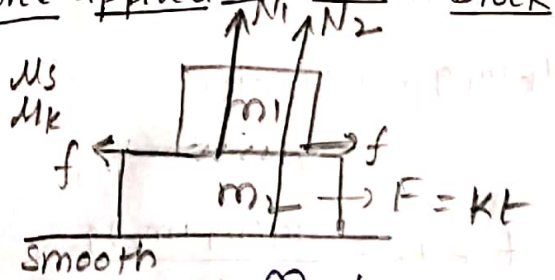
$mg\sin\theta - \sqrt{2}\mu mg\cos\theta = ma$

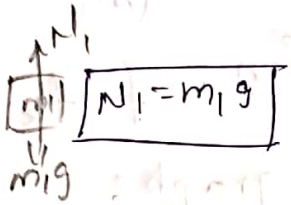
$$\boxed{g(\sin\theta - \sqrt{2}\mu\cos\theta) = a}$$

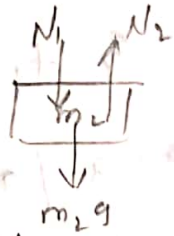


Block on Block problems:

Force applied on lower block:







$N_2 = N_1 + m_2 g$

$= m_1 g + m_2 g$

$$\boxed{N_2 = (m_1 + m_2)g}$$

(q) $N_1 = m_1 g$, $N_2 = (m_1 + m_2)g$

(q) $f_s = \mu_s m_1 g$

$f_k = \mu_k N_1 = \mu_k m_1 g$

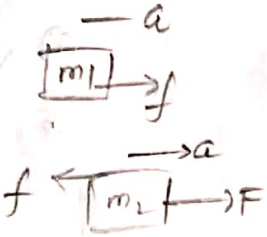
* Let the blocks move together with common acceleration 'a'.

$$a_1 = a_2 = a = \frac{F}{m_1 + m_2} = \frac{kt}{m_1 + m_2}$$

now, static friction blw the blocks,

$f = m_1 a$

$$\boxed{f = \frac{m_1 kt}{m_1 + m_2}}$$



we have,

$f \leq f_s$

$$\frac{m_1 kt}{m_1 + m_2} \leq \mu_s m_1 g \Rightarrow \boxed{t \leq \frac{\mu_s g(m_1 + m_2)}{k}}$$

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* when $t > \frac{\mu_s(m_1+m_2)g}{k}$, block move with different accelerations.

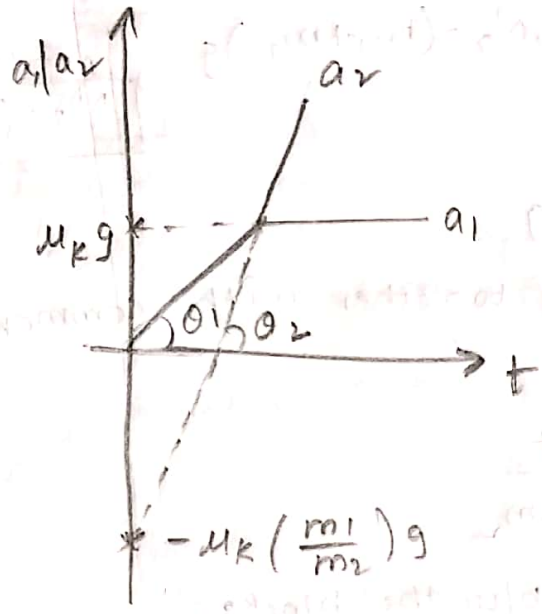
$$a_1 = \frac{f_k = \mu_k m_1 g}{m_1} = \frac{\mu_k g}{1} = \mu_k g - \text{const}.$$

$$a_2 = \frac{F - f_k}{m_2} = \frac{F - \mu_k m_1 g}{m_2}$$

$$a_2 = \left(\frac{k}{m_2}\right)t - \mu_k\left(\frac{m_1}{m_2}\right)g$$

& friction, $f = f_k = \mu_k m_1 g \rightarrow$ Kinetic.

* a_1 / a_2 - graph & $\hookrightarrow$ a vst.



$$t = t_0 = \frac{\mu_s(m_1+m_2)g}{k}$$

For $t < t_0$;

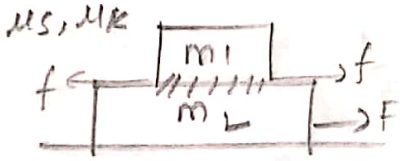
$$a_1 = a_2 = \frac{kt}{m_1+m_2}$$

$$\tan\theta_1 = \frac{k}{m_1+m_2}$$

$t > t_0$;

$$a_1 = \mu_k g$$

$$a_2 = \frac{k}{m_2}t - \mu_k\left(\frac{m_1}{m_2}\right)g$$



① $N_1 = m_1 g$; $N_2 = (m_1+m_2)g$

② $f_s = \mu_s m_1 g$

$f_k = \mu_k m_1 g$

* let blocks move with common acceleration

$$a_1 = a_2 = a = \frac{F}{m_1+m_2}$$

* Static friction b/w the blocks,

$$f = m_1 a = \boxed{\frac{m_1 F}{m_1+m_2} = f}$$

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we have, $f \le f_s$

$$\frac{m_1 F}{m_1+m_2} \le \mu_s m_1 g$$

$$\boxed{F \le \mu_s (m_1+m_2) g}$$

maximum force upto which blocks move together,

$$\boxed{F_{max} = \mu_s (m_1+m_2) g} \quad **$$

* If $F \le F_{max}$, blocks move with same acceleration,

$$a_1 = a_2 = \frac{F}{m_1+m_2}, \quad f = \frac{m_1 F}{m_1+m_2}$$

$a_{max} \Rightarrow \frac{f_k}{m_1} = \frac{\mu_k m_1 g}{m_1} = \mu_k g \rightarrow$ static. $\rightarrow$ static.

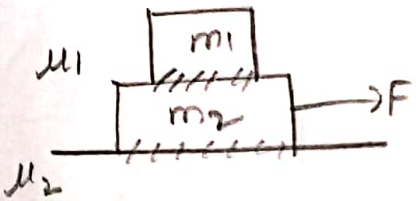
* If $F > F_{max}$, blocks move with diff acceleration,

$$a_1 = \frac{f_k}{m_1} = \frac{\mu_k m_1 g}{m_1} = \mu_k g$$

$$a_2 = \frac{F - f_k}{m_2} = \frac{F - \mu_k m_1 g}{m_2}$$

friction $\rightarrow$ kinetic friction & $f_k = \mu_k m_1 g$

*



① $N_1 = m_1 g$, $N_2 = (m_1+m_2) g$

$f_{1l} = \mu_1 N_1 = \mu_1 m_1 g$

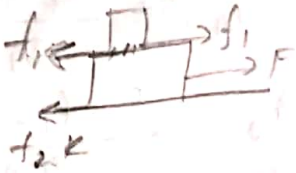
$f_{2l} = \mu_2 (m_1+m_2) g$

* Slipping first occurs blw lower block & ground.

* If $F \le f_{2l}$; then blocks are at rest. $a_1 = a_2 = 0$

* If $F > f_{2l}$; let blocks move together with common acceleration ('a').

$$a_1 = a_2 = a = \frac{F - f_{2k}}{m_1+m_2}$$



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$$a = \frac{F - \mu_2 (m_1+m_2) g}{m_1+m_2}$$

* Static friction b/w blocks,

$$f_1 = m_1 a$$

$$f_1 = \frac{m_1 [F - \mu_2 (m_1+m_2) g]}{m_1+m_2}$$

we have, $f_1 \le f_{1l}$

$$\frac{m_1 [F - \mu_2 (m_1+m_2) g]}{m_1+m_2} \le \mu_1 m_1 g$$

$$F \le (\mu_1+\mu_2)(m_1+m_2) g$$

max force upto which blocks move together

$$F_{max} = (\mu_1+\mu_2)(m_1+m_2) g$$

** If $F \le f_{2l}$ then $a_1 = a_2 = 0$ & $f_2 = F$; $f_1 = 0$ (static)

* If $F > f_{2l}$ & $F \le F_{max}$

then $a_1 = a_2 = a = \frac{F - f_{2k}}{m_1+m_2}$

$f_2 = f_{2k} = \mu_2 (m_1+m_2) g \rightarrow$ kinetic

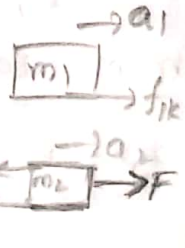
$f_1 = m_1 a$; static.

* If $F > f_{2l}$ & $F > F_{max}$

block move with seperate accelerations.

$$a_1 = \frac{f_{1k}}{m_1} = \frac{\mu_1 m_1 g}{m_1} = \mu_1 g$$

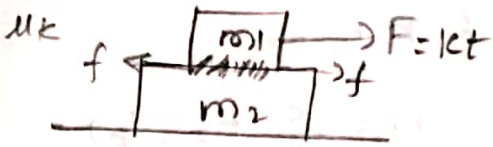
& $a_2 = \frac{F - f_{1k} - f_{2k}}{m_2}$



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& frictions :- $f_1 = f_{1k}$, $f_2 = f_{2k}$.

Force applied on upper block :-



① $N_1 = m_1 g$, $N_2 = (m_1 + m_2) g$

② $f_s = \mu_s N_1 = \mu_s m_1 g$

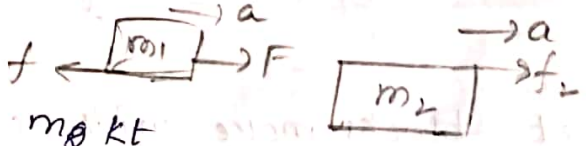
$f_k = \mu_k N_1 = \mu_k m_1 g$

* Let blocks move together with common acceleration

$$a_1 = a_2 = a = \frac{F}{m_1 + m_2} = \frac{kt}{m_1 + m_2}$$

* Static friction b/w the blocks,

$\therefore f = m_2 a$



$$f = \frac{m_2 F}{m_1 + m_2} = \frac{m_2 kt}{m_1 + m_2}$$

we have,

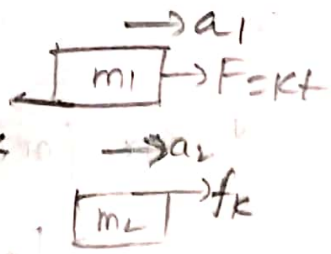
$$f \leq f_s$$

$$\frac{m_2 kt}{m_1 + m_2} \leq \mu m_1 g$$

$$\boxed{t \leq \frac{\mu \left(\frac{m_1}{m_2}\right) (m_1 + m_2) g}{k}}$$

* for $t > t_0$; blocks move with diff accelerations.

$$a_1 = \frac{kt - f_k}{m_1} = \frac{kt - \mu_k m_1 g}{m_1}$$

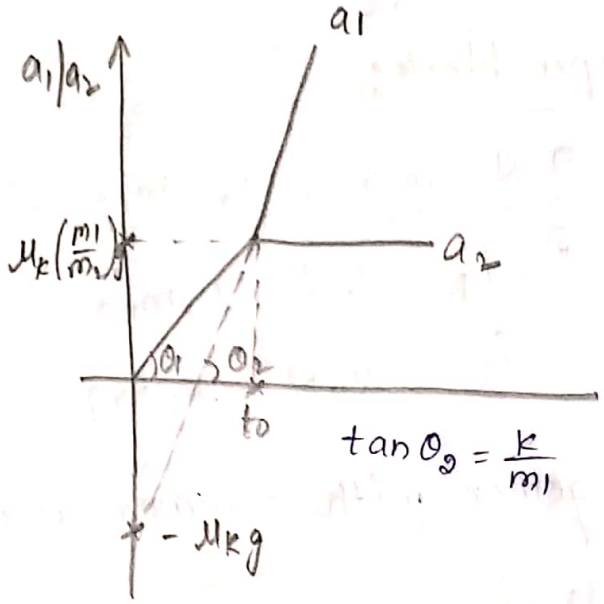


$$\boxed{a_1 = \left(\frac{k}{m_1}\right) t - \mu_k g}$$

& $a_2 = \frac{f_k}{m_2} = \frac{\mu_k m_1 g}{m_2} = \text{const.}$

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a_1/a_2 vs t -graph:



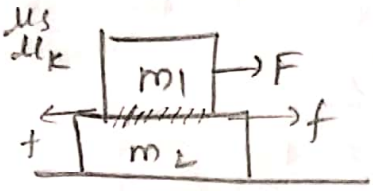
For $t < t_0$

$$a_1 = a_2 = \frac{kt}{m_1+m_2}$$

For $t > t_0$

$$a_1 = \frac{k}{m_1}t = \mu_k g$$

$$a_2 = \mu_k\left(\frac{m_1}{m_2}\right)g$$



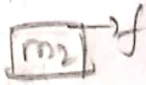
① $N_1 = m_1 g$, $N_2 = (m_1+m_2)g$

② $f_s = \mu_s m_1 g$, $f_k = \mu_k m g$

* Let blocks move with common acceleration a.

$$a_1 = a_2 = a = \frac{F}{m_1+m_2}$$

& static friction b/w blocks,



$$f = m_2 a \Rightarrow \boxed{f = \frac{m_2 F}{m_1+m_2}}$$

we have,

$$f \le f_s$$

$$\frac{m_2 F}{m_1+m_2} \le \mu_s m_1 g$$

$$\boxed{F \le \mu_s\left(\frac{m_1}{m_2}\right)(m_1+m_2)g}$$

* max force upto which blocks move together,

$$\boxed{F_{max} = \mu_s\left(\frac{m_1}{m_2}\right)(m_1+m_2)g}$$

* If $F \le F_{max}$,

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$a_1 = a_2 = \frac{F}{m_1+m_2}$ & $f = m_2 a$ (static)

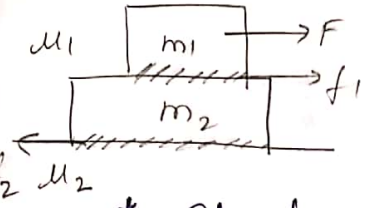
* If $F > F_{max}$ then blocks moves with different accelerations.

$$a_1 = \frac{F - f_k}{m_1} = \frac{F - \mu_k m_1 g}{m_1}$$

$$a_2 = \frac{f_k}{m_2} = \frac{\mu_k m_1 g}{m_2}$$

& $f = f_k = \mu_k m_1 g \longrightarrow$ kinetic

*



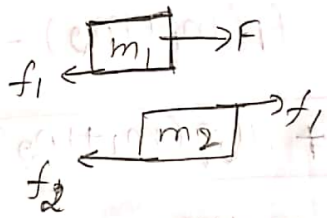
① $N_1 = m_1 g$, $N_2 = (m_1+m_2)g$

② $f_{1l} = \mu_1 m_1 g$, $f_{2l} = \mu_2 (m_1+m_2) g$

* If $f_{1l} < f_{2l}$; lower block is at rest & here if $F < F_{1l}$ upper block is also at rest.

$F - f_1 = m(0)$

$\boxed{F = f_1}$, static



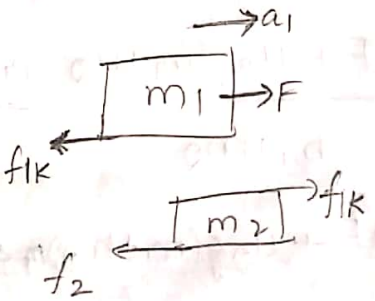
& $f_1 - f_2 = m_2(0)$

$\boxed{f_1 = f_2}$

* $F > f_{1l}$ & $f_{1l} < f_{2l}$.

$F - f_{1k} = m_1 a_1$

$$\boxed{a_1 = \frac{F - f_{1k}}{m_1}}$$



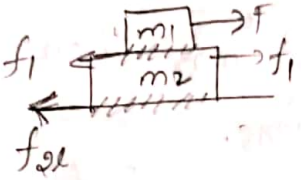
& $\boxed{a_2 = 0} \Rightarrow f_2 = f_{1k}$, static

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*If $f_{1l} > f_{2l}$ & $F > f_{2l}$

Blocks are in motion

[Here slipping takes place b/w bottom block & ground]



* Let both the blocks move with common acceleration

$$a_1 = a_2 = a = \frac{F - f_{2K}}{m_1 + m_2} = \frac{F - \mu_2 (m_1 + m_2) g}{m_1 + m_2}$$

* static friction b/w the blocks

$$a = \frac{F - f_1}{m_1}$$

$$\frac{F - \mu_2 (m_1 + m_2) g}{m_1 + m_2} = \frac{F - f_1}{m_1}$$

$$m_1 [F - \mu_2 (m_1 + m_2) g] = (F - f_1)(m_1 + m_2)$$

$$m_2 F = f_1 (m_1 + m_2) - \mu_2 g (m_1 + m_2) m_1$$

$$\boxed{\frac{m_2 F + \mu_2 g (m_1 + m_2) m_1}{m_1 + m_2} = F}$$

In general,

$$f_1 \leq f_{1l}$$

$$\frac{m_2 F + \mu_2 g (m_1 + m_2) m_1}{m_1 + m_2} \leq \mu_1 m_1 g$$

$$m_2 F + \mu_2 g (m_1 + m_2) m_1 \leq \mu_1 m_1 g (m_1 + m_2)$$

$$m_2 F \leq (m_1 + m_2)(\mu_1 m_1 g - \mu_2 g) m_1$$

$$\boxed{F \leq \frac{m_1 (m_1 + m_2) g (\mu_1 m_1 - \mu_2)}{m_2}}$$

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$$F_{max} = \frac{m_1}{m_2}(m_1+m_2)(\mu_1-\mu_2)g$$

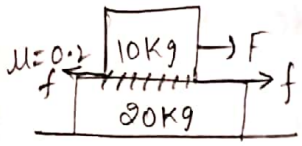
* If $F > F_{max}$; block move with different accelerations.

$$a_1 = \frac{F - f_{1k}}{m_1}$$

$$a_2 = \frac{f_{1k} - f_{2k}}{m_2}$$

frictions: $f_1 = f_{1k} = \mu_1 m_1 g$; kinetic

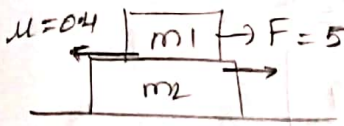
$f_2 = f_{2K} = \mu_2(m_1+m_2)g$; kinetic.



$$a = \frac{F}{30}$$

$$F_{max} = 0.2 \times \frac{20}{10}(30)(10) \quad 0.2 \times \frac{10}{20} \times 30 \times 10$$

$$F_{max} = 30\,N$$



$m_1 = 2kg$

$m_2 = 5kg$

$$F_{max} = 0.4 \times \frac{2}{5} \times 7 \times 10$$

$$= 11.2\,N$$

$\Rightarrow F > F_{max}$

different accelerations

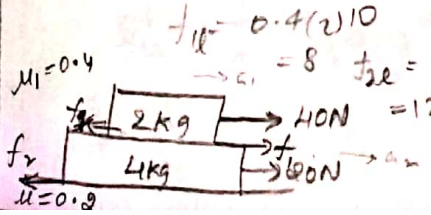
$$a_1 = \frac{50 - 0.4 \times 2 \times 10}{2} = 25 - 4 = 21\ m/s^2$$

$$a_2 = \frac{0.4 \times 2 \times 10}{8} = 1.6\ m/s^2$$

$f_{1k} = 0.4(2)10$

$= 8$ $f_{2k} = 0.2(6)10$

$= 12$

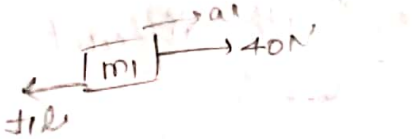


$$a = \frac{60-12}{6} = \frac{48}{6} = 8\ m/s^2$$

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$40 - f_1 = 2 \times 8$

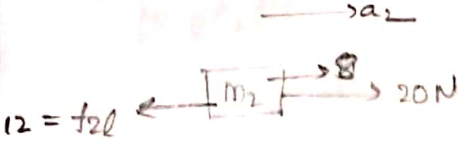
$\boxed{f_1 = 24\,N}$ ✗



$40 - 16 = a_1(2)$

$32 = a_1(2)$

$a_1 = 16\ m/s^2$

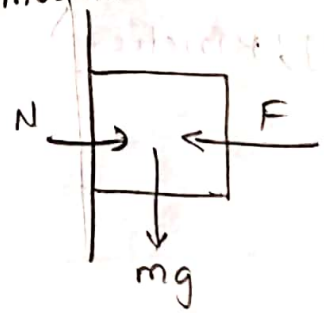


$a_2 = \frac{28-12}{4} = \frac{16}{4}$

$= 4\ m/s^2$ //

Block in contact with Vertical Surface

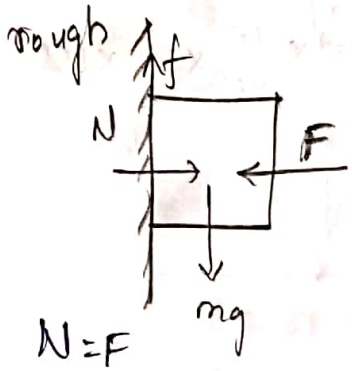
Smooth



Block cannot stay in equilibrium

& $a = \frac{mg}{m} = g$

Rough



Block is at rest, when $mg \le f_l$.

$\Rightarrow mg \le \mu_s N$

$\frac{mg}{\mu_s} \le F$

$\boxed{F_{min} = \frac{mg}{\mu_s}}$

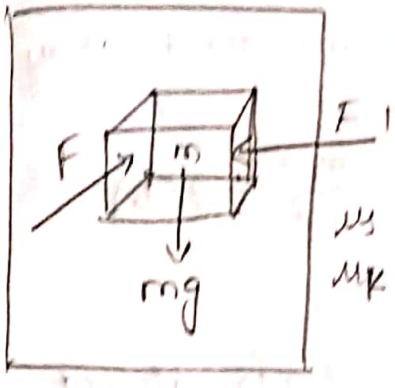
* If $F < F_{min}$ (or) $mg > f_l$, block is in motion.

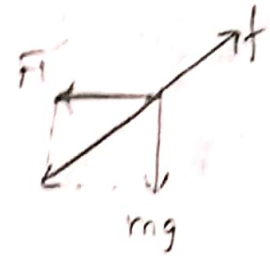
$\therefore a = \frac{mg - f_k}{m} = \frac{mg - \mu_k N}{m} \Rightarrow \boxed{a = \frac{mg - \mu_k F}{m}}$

* If $F > F_{min}$ (or) $mg < f_l$, block is at rest.

$\boxed{a = 0}$ & $\boxed{f_s = mg}$

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$F_R = \sqrt{F_f^2 + (mg)^2}$

* If $F_R \le f_s$ then block is at rest.

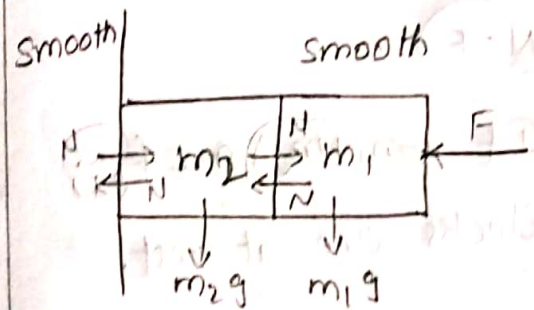
$\therefore$ $\boxed{a = 0}$ & $F_s = F$.

* If $F_R > f_s$ then block is in motion,

$$\boxed{a = \frac{F_R - f_k}{m}} \Rightarrow f_k = \mu_k N = \mu_k F$$

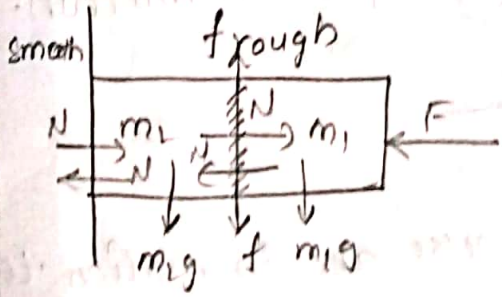
& friction, $f_k = \mu_k N = \mu_k F \rightarrow$ kinetic)

* Block on Block in contact with Vertical Surface



$$a_1 = \frac{m_1 g}{m_1} = g(\downarrow)$$

$$a_2 = \frac{m_2 g}{m_2} = g(\downarrow)$$

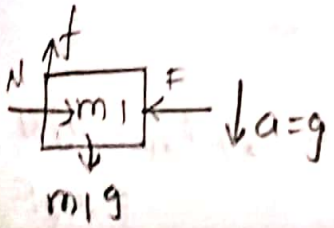


$N = F$

* let the system moves down with common acceleration 'a'.

$$a_1 = a_2 = a = \frac{(m_1 + m_2)g}{m_1 + m_2} = g(\downarrow)$$

* Static friction b/w the blocks —



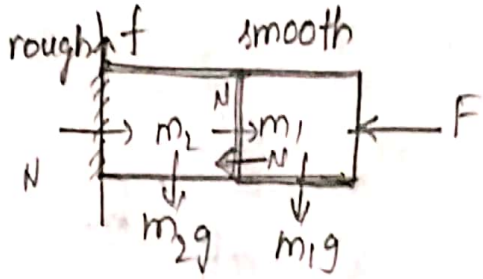
$m_1 g - f = m_1 a$

$m_1 g - f = m_1 g$ our assumption is correct $\Rightarrow \boxed{a_1 = a_2 = g}$

$\boxed{f = 0}$

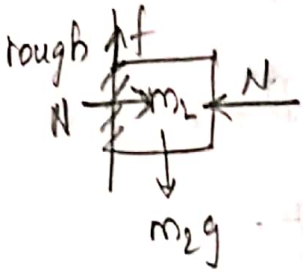
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*



clearly, m_1 cannot stay at rest.

* $a_1 = \frac{m_1 g}{m_1} = g(\downarrow)$



$m_2 g \le f_l$ m_2 is at rest.

$m_2 g \le \mu_s F$

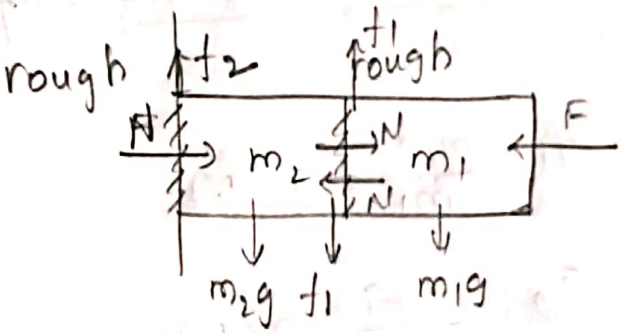
$$F \ge \frac{m_2 g}{\mu_s}$$

* If $F < \frac{m_2 g}{\mu_s}$; blocks in motion.

$$a = \frac{m_2 g - f_k}{m_2} = \frac{m_2 g - \mu_k F}{m_2} \ (\downarrow)$$

Eq $f_k = \mu_k F$ → Kinetic.

*



$N = F$

* If $(m_1+m_2)g \le f_{2l}$; Blocks are at rest.

and here if $m_1 g > f_{1l}$; m_1 is in motion.

$$a_1 = \frac{m_1 g - f_{1k}}{m_1}$$

* If $(m_1+m_2)g > f_{2l}$; blocks are in motion. Let us assume blocks move with common acceleration.

$$a_1 = a_2 = a = \frac{(m_1+m_2)g - f_{2k}}{m_1+m_2}$$

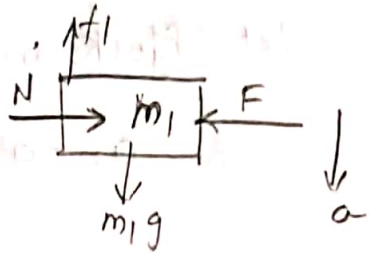
$f_{2k} = \mu_2 N = \mu_2 F$.

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* static friction b/w the blocks :

$m_1 g - f_1 = m_1 a$

from this we get 'f_1'



* If $f_1 \le f_{1s}$ our assumption is correct.

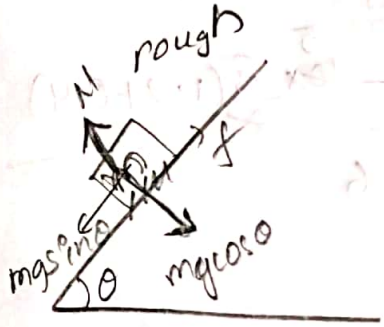
i.e, $\boxed{a_1 = a_2 = a}$

* If $f_1 > f_{1s}$ then blocks move with different accelerations.

$a_1 = \frac{m_1 g - f_{1K}}{m_1}$ & $a_2 = \frac{m_2 g + f_{1K} - f_{2K}}{m_2}$.

* Block placed on inclined plane

*



** Block is at rest when $mg\sin\theta \le f_s$

$mg\sin\theta \le mg\cos\theta \cdot \mu_s$

$\tan\theta \le \mu_s$

$\boxed{\mu_s \ge \tan\theta}$

Here friction $f = mg\sin\theta$.

* If $\mu < \tan\theta$, Block slides down with acceleration 'a'.

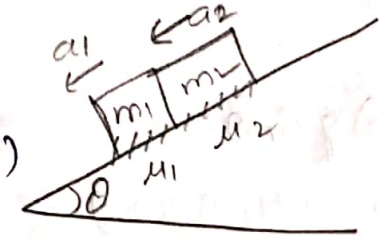
$a = \frac{mg\sin\theta - f_K}{m} = \frac{mg(\sin\theta - \mu_K\cos\theta)}{m}$

$\boxed{a = g(\sin\theta - \mu_K\cos\theta)}$

→ Find the contact force b/w the blocks

① $a_1 = g(\sin\theta - \mu_1\cos\theta)$

$a_2 = g(\sin\theta - \mu_2\cos\theta)$



* If $a_1 = 0$ (or) $a_1 > a_2$ $\Rightarrow \boxed{N = 0}$

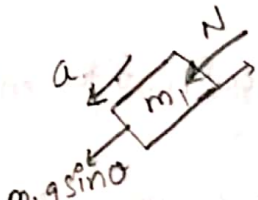
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***If $a_1 < a_2$ then $N \neq 0$

Let blocks move with common accelerations.

$$a_1 = a_2 = \frac{(m_1+m_2)g\sin\theta - (f_1+f_2)}{m_1+m_2}$$

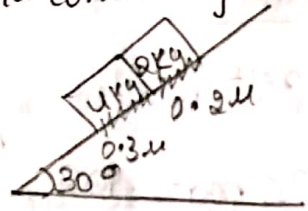
$$a = \frac{(m_1+m_2)g\sin\theta - g\cos\theta(\mu_1 m_1 + \mu_2 m_2)}{m_1+m_2}$$



$$m_1 g\sin\theta + N - \mu_1 m_1 g\cos\theta = m_1 a$$

⇒ from this N can be found.

Find contact force b/w the blocks.



$$a = \frac{6(10)\times\frac{1}{2} - 10\times\frac{\sqrt{3}}{2}(1.2+0.4)}{6}$$

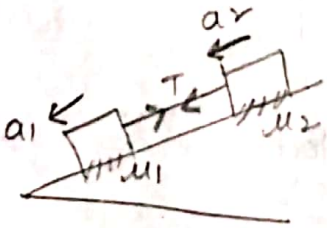
$$= \frac{30-8\sqrt{3}}{6}$$

$$4\times10\times\frac{1}{2} + N - 0.3\times4\times10\times\frac{\sqrt{3}}{2} = \frac{2(30-6\sqrt{3})}{3}$$

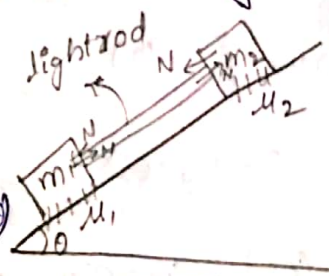
$$20 + N - 60\sqrt{3} = 20 - 16\frac{\sqrt{3}}{3}$$

$$N = 60\sqrt{3} - 16\frac{\sqrt{3}}{3} = \sqrt{3}\left(\frac{120-16}{3}\right)$$

$$N = \frac{104}{\sqrt{3}}$$



Tension in string.



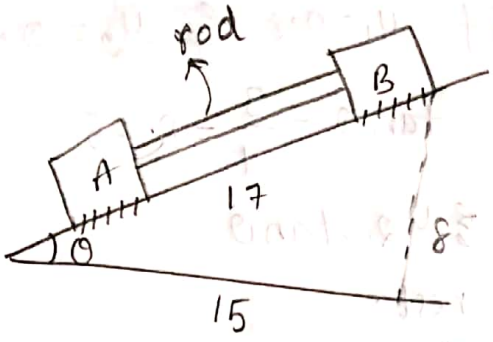
*If $(m_1+m_2)g\sin\theta \leq (\mu_1 m_1 g\cos\theta + \mu_2 m_2 g\cos\theta)$ then blocks are at rest $\therefore \boxed{N=0}$

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* If $(m_1+m_2)g\sin\theta > (\mu_1 m_1 g\cos\theta + \mu_2 m_2 g\cos\theta)$

then $N \neq 0$ $\quad a_1 = a_2 = ?$

↳ common acceleration.



$m_A = m_B = 170\,kg$

$\mu_A = 0.2$, $\mu_B = 0.4$,

$g = 10\,m/s^2$

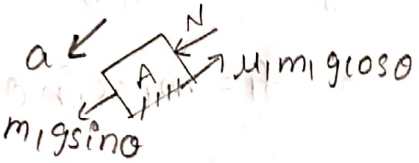
Find force developed in the rod. (150 N)

$$(m_1+m_2)g\sin\theta = (\cancel{340}^{20})(10)\left(\frac{8}{\cancel{17}}\right) = 1600$$

$$(\mu_1 m_1 + \mu_2 m_2)g\cos\theta = \cancel{170}^{10}(0.6)(10)\left(\frac{15}{\cancel{17}}\right) = 900$$

$1600 > 900$

$\Rightarrow a_1 = a_2$



$$m_1 g\sin\theta + N - \mu_1 m_1 g\cos\theta = m_1 a$$

$$\cancel{170}^{10}\times 10\times\frac{8}{\cancel{17}} + N - 0.2\times\cancel{170}^{10}\times 10\times\frac{15}{\cancel{17}} = 170(a)$$

$$800 + N - 300 = 170(a)$$

$$500 + N = 170(a)$$

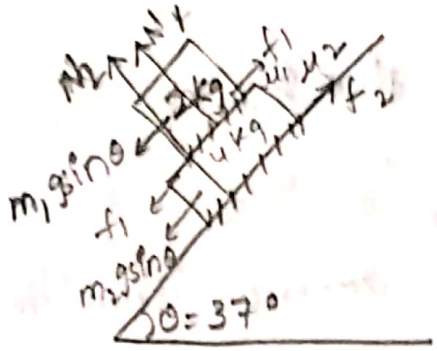
$$N = 170(a) - 500$$

$$a = \frac{\cancel{340}^{20}\times 10\times\frac{8}{\cancel{17}} - 10\times\frac{15}{\cancel{17}}\times\cancel{170}^{10}\times 0.6}{340}$$

$$= \frac{1600-900}{340} = \frac{700}{340} = \frac{70}{34}$$

$$N = \cancel{170}^{\cancel{10}5}\times\frac{70}{\cancel{34}_{\cancel{2}}} - 500 = 350 - 500 = 150$$

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$N_1 = m_1 g\cos\theta = 2 \times 10 \times \frac{4}{5} = 16N$

$N_2 = 6 \times 10 \times \frac{4}{5} = 48N$

① If $\mu_1 = 0.8$ & $\mu_2 = 0.8$

we have $\tan\theta = \frac{3}{4} = 0.75$

$\mu_1 > \tan\theta$ & $\mu_2 > \tan\theta$

∴ Blocks are at rest.

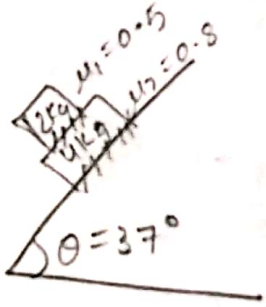
*friction b/w Block B & inclined plane.

$f_2 = f_1 + m_2 g\sin\theta = (m_1 + m_2) g\sin\theta$

$= 6 \times 10 \times \frac{3}{5} = 36N \rightarrow$ static

* friction b/w Blocks

$f_1 = m_1 g\sin\theta = 2 \times 10 \times \frac{3}{5} = 12N \rightarrow$ static.

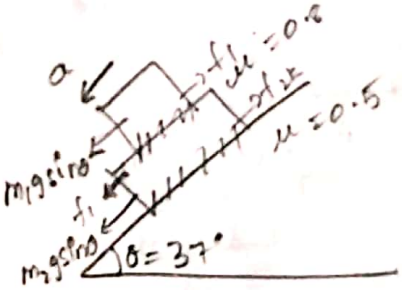


lower block is at rest $(a_2 = 0)$

upper block is in motion.

$a_1 = \frac{m_1 g\sin\theta - f_{1k}}{m_1}$ & $f_1 = f_{1k}$

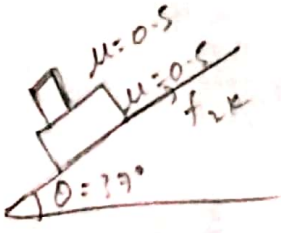
$f_2 = f_{1k} + m_2 g\sin\theta$



* $a_1 = a_2 = a = \frac{(m_1 + m_2) g\sin\theta - f_{2k}}{m_1 + m_2}$

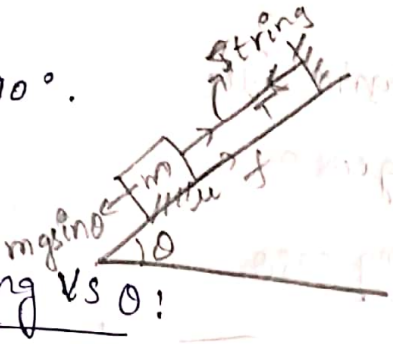
* static friction b/w the blocks

$m_1 g\sin\theta - f_1 = m_1 a$



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θ changes θ = 0° to 90°.



graphs!

① Tension in the string Vs θ:

$mg\sin\theta \le f_s \Rightarrow T = 0$

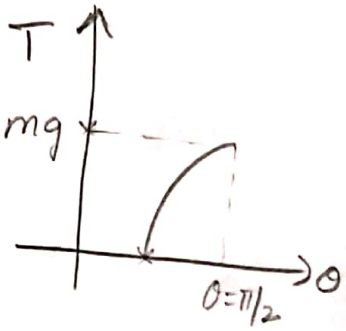
$mg\sin\theta \le \mu_s mg\cos\theta$

$\boxed{\tan\theta \le \mu}$

when $mg\sin\theta > f_s$ then $T \neq 0$

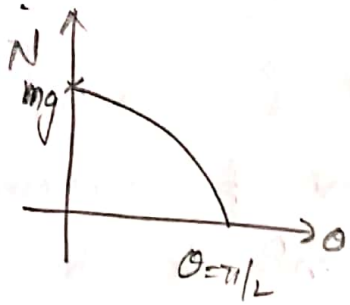
$T = mg\sin\theta - \mu mg\cos\theta$

$\boxed{T = mg(\sin\theta - \mu\cos\theta)}$



② Normal reaction Vs θ:

$N = mg\cos\theta$



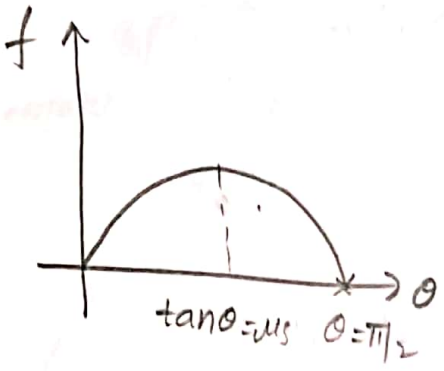
③ friction force blw block & incline Vs θ:

$\tan\theta \le \mu$

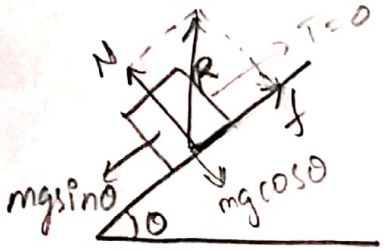
$f = mg\sin\theta$

& $\tan\theta > \mu$

$f = \mu mg\cos\theta$



④ Net contact force blw block & incline Vs θ:



when $\tan\theta \le \mu_s$

$N = mg\cos\theta$, $f = mg\sin\theta$

$\boxed{R = mg}$

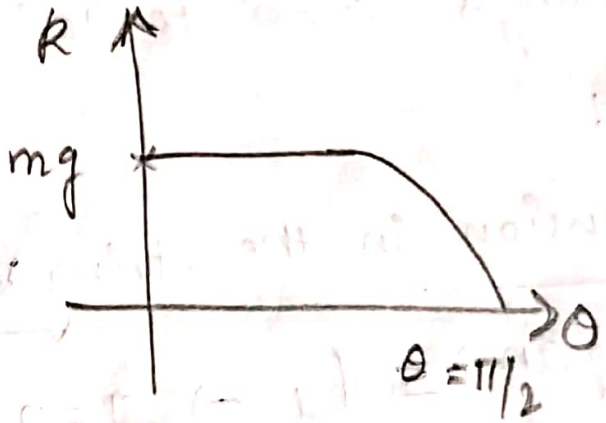
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when $\tan\theta > \mu_s$

$N = mg\cos\theta$

$f = \mu mg\cos\theta$

$$\boxed{R = mg\cos\theta\sqrt{1+\mu^2}}$$



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